Talagrand's two-cost discrepancy problem

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Abstract

This note addresses Research Problem 17.2.2 in the second edition of M. Talagrand's *Upper and Lower Bounds for Stochastic Processes: Decomposition Theorems*. We give a necessary and sufficient condition for the discrepancy bound for every pair of finite convex Young costs. With $R_i(j) = (\theta_i^*)^{-1}(\exp(2^j))$, the condition is that, for some $c > 0$, both prefix sums $\sum_{j\le J} \exp[-cR_1(j)^2 R_2(J)^2/2^j]$ and their coordinate interchanges remain bounded uniformly in J. The proof balances the two coordinate costs across scales through an adaptive Gaussian construction, with matching lower bounds obtained from Brownian local time. A nonmeasurable counterexample explains why completely unrestricted functions cannot simply be replaced by their convex envelopes. We thank M. Talagrand for pointing us towards this technical problem as a way to gain better insight on the Ultimate Matching Problem.

1 Statements and the scope of the answer

Let $n = 2^p$ and $G_n = \{1, \dots, n\}^2$. Differences D_i are taken only over internal edges. Put

$$\mathcal{H}_n(\theta_1, \theta_2) = \left\{ h : G_n \to \mathbb{R} : \sum \theta_1(|D_1 h|) + \sum \theta_2(|D_2 h|) \le n^2 \right\}.$$

For independent standard normal variables g_v define

$$w_n = \mathbb{E} \sup_{h \in \mathcal{H}_n} \left| \sum_{v \in G_n} g_v (h(v) - \bar{h}) \right|, \qquad \bar{h} = n^{-2} \sum_v h(v).$$

The empirical setting is the one in the book: integers $m_0 \le n(v) \le 2m_0$, $N = \sum_v n(v)$, $\mu(v) = n(v)/N$, and independent U_i of law μ. The relevant density regime is $m_0 \ge c_0 p$, where $c_0 > 0$ is fixed. All constants below are uniform in p, m_0, and the admissible counts; they may depend on the two costs and on c_0. The discrepancy of an empty admissible class is understood as zero.

Theorem 1.1 (Two Young costs)**.** *Suppose $\theta_i : [0,\infty) \to [0,\infty)$ are finite, convex, $\theta_i(0) = 0$, and $\theta_i(t) \ge t$. Let $\psi_i = \theta_i^*$ and let $\rho_i = \psi_i^{-1}$ be the increasing generalized inverses. Put $R_i(j) = \rho_i(\exp(2^j))$. The following are equivalent:*

1. *$w_{2^p} \le C 2^{2p} \sqrt{p}$ for every $p \ge 1$.*

2. For some $c > 0$,

$$\sup_{J\ge 1}\sum_{j=1}^{J}\exp\left[-c\frac{R_1(j)^2R_2(J)^2}{2^j}\right]<\infty,\qquad \sup_{J\ge 1}\sum_{j=1}^{J}\exp\left[-c\frac{R_2(j)^2R_1(J)^2}{2^j}\right]<\infty. \tag{1.1}$$

3. The estimate (17.18) holds uniformly in the above empirical setting:

$$\mathbb{E}\sup_{h\in\mathcal{H}_n}\left|\sum_{i=1}^{N}(h(U_i)-\mu h)\right|\le C\sqrt{pm_0}\,n^2.$$

The same criterion results on replacing $R_i(j)$ by $e^{2^j}/\theta_i^{-1}(e^{2^j})$. In (2), the two suprema can equivalently both be required to be at most e^{-1} after increasing c.

The condition is symmetric. It includes growth profiles whose relative sizes change infinitely often; no regular variation or common preferred coordinate is assumed. Its prefix structure is essential. The simpler pointwise condition $R_1(J)R_2(J)\gtrsim\sqrt{2^J}$ is necessary but not sufficient.

Theorem 1.2 (Finite Borel costs)**.** *Let $\theta_i:[0,\infty)\to[0,\infty)$ be finite Borel functions with $\theta_i(t)\ge t$. Let ϕ_i be the greatest convex minorant on $\mathbb{R}$ of $s\mapsto\theta_i(|s|)$, and put*

$$a_i=\inf\theta_i=\phi_i(0),\qquad a=a_1+a_2,\qquad \vartheta_i(t)=\phi_i(t)-a_i+t.$$

If $a\ge 1$, the desired estimate always holds. If $a<1$, it holds if and only if (1.1) *holds for the normalized Young pair $(\vartheta_1,\vartheta_2)$.*

Every finite nondecreasing cost is Borel and locally bounded, so this covers arbitrary pairs in the usual monotone-cost interpretation of the book's question. The Borel assumption can also be replaced by finite Lebesgue measurability, by completing the finite product measures in the exceptional-edge argument. A cost that is locally bounded need not be measurable at all; the same classification still applies, as explained in the relaxation proof. Completely unrestricted functions require care: the final section gives an explicit pair for which separate convex envelopes give the wrong answer. We do not claim a classification of all such nonmeasurable pairs.

1.1 Reading the criterion

If $\theta_1(t)=t$, then $R_1(j)=1$ and the theorem reduces exactly to

$$\exists c>0:\qquad \sum_{j\ge 1}\exp\left[-c\frac{R_2(j)^2}{2^j}\right]<\infty.$$

Equivalently,

$$\exists c>0:\qquad \int_{e^e}^{\infty}\exp\left[-c\frac{s^2}{\theta_2^{-1}(s)^2\log s}\right]\frac{ds}{s\log s}<\infty.$$

Indeed, the first prefix condition has geometrically decaying tails once $R_2(J)^2/2^J$ is bounded below; the second becomes the displayed infinite series. The inverse-product inequality below and dyadic integration give the integral form. Consequently the sharp borderline example in this direction is $t\sqrt{\log t\,\log\log\log t}$; $t\sqrt{\log t}$ fails. For equal normalized Young costs the exact condition is simply $\theta(t)\gtrsim t(\log t)^{1/4}$. Further exact examples are proved after the main argument.

Lemma 1.3 (Complementary inverses)**.** *For a finite convex Young function $\theta \geq t$ and its complement ψ, at every $v > 0$,*

$$v \leq \theta^{-1}(v)\psi^{-1}(v) \leq 2v.$$

Proof. Young's inequality gives the upper bound. For the lower bound set $x = \theta^{-1}(v)$ and $y = v/x$. If $z \leq x$, then $yz - \theta(z) \leq v$. If $z \geq x$, convexity gives $\theta(z)/z \geq \theta(x)/x = y$. Thus $\psi(y) \leq v$ and $\psi^{-1}(v) \geq y$. Limits give the same assertions for generalized inverses when the complement has a zero interval or is extended valued. □

1.2 Conventions and elementary Gaussian comparisons

On the unit square, W is the isonormal Gaussian process on L^2: $\mathbb{E}W(f)W(g) = \int fg$. Write $W^0(f) = W(f - \int f)$. If P_h is cell averaging on the dyadic h-square grid, its histogram density has value $h^{-2}W^0(Q)$ on each cell Q. A zero-normal current of divergence f means an integrable vector field J such that $\int f\varphi = -\int J \cdot \nabla\varphi$ for smooth tests on the closed square. For a convex complementary cost ψ, use the Luxemburg norm

$$\|g\|_{L^\psi} = \inf\{r > 0 : \int \psi(|g|/r) \leq 1\}.$$

It is a norm, including the extended-valued case: convexity proves the triangle inequality by taking the convex combination with weights $r/(r+s)$ and $s/(r+s)$. Young's pointwise inequality gives $\int |fg| \leq 2\|f\|_{L^\theta}\|g\|_{L^\psi}$.

Lemma 1.4 (Gaussian comparison)**.** *If two finite centered Gaussian processes satisfy $\mathbb{E}(X_i - X_j)^2 \leq \mathbb{E}(Y_i - Y_j)^2$ for every i, j, then $\mathbb{E}\max_i X_i \leq \mathbb{E}\max_i Y_i$. A centered Gaussian covariance dominated by another has smaller expectation for every convex function for which the expectations exist. Finally, for deterministic vectors a_i and a convex positively homogeneous function F,*

$$\mathbb{E}_\varepsilon F\left(\sum_i \varepsilon_i a_i\right) \leq \sqrt{\pi/2}\,\mathbb{E}_g F\left(\sum_i g_i a_i\right).$$

Proof. For the first assertion interpolate independent processes by $Z_t = \sqrt{t}X + \sqrt{1-t}Y$, and apply Gaussian integration by parts to $f(z) = \beta^{-1}\log\sum_i e^{\beta z_i}$. Writing $w_i = e^{\beta z_i}/\sum_j e^{\beta z_j}$, its derivative is

$$\frac{d}{dt}\mathbb{E}f(Z_t) = \frac{\beta}{4}\mathbb{E}\sum_{i,j} w_i w_j\{\mathbb{E}(X_i - X_j)^2 - \mathbb{E}(Y_i - Y_j)^2\} \leq 0.$$

Let $\beta \to \infty$. Infinite compact index sets follow by finite approximations. For covariance domination, add an independent centered Gaussian of the covariance difference and use conditional Jensen. For the last assertion write $g_i = \varepsilon_i|g_i|$, and condition on their independent signs. Jensen over the absolute values gives the right side at least $\mathbb{E}|g_1|$ times the left side. □

Symmetrization used below also follows directly from Jensen: introduce independent observations U_i', bound the original centered support function by its difference-sample expectation, exchange each pair using an independent sign, and apply the triangle inequality. The result is twice the signed-process expectation. Subtracting μh from every test before this operation removes the otherwise unbounded constants.

The one-dimensional Gaussian bounds used throughout are $\mathbb{P}(|g| \geq t) \leq 2e^{-t^2/2}$ and $\mathbb{P}(|g| \geq t) \geq ce^{-t^2}$ for $t \geq 0$. For the second, integrate the Gaussian density from t to $t + (1+t)^{-1}$ and adjust the fixed constant on $[0, 1]$. Independence then gives $\mathbb{E}\max_{j\leq M}|g_j| \geq c\sqrt{\log(M+1)}$;

a union bound gives the matching upper estimate. Stop-loss comparisons imply convex order because a convex piecewise linear function is an affine function plus a nonnegative combination of $(x-t)_+$; approximation extends this to convex Lipschitz functions. All support functions used in the finite-grid comparisons are of this type.

2 Gaussian, continuum, and empirical reductions

2.1 Grid and continuum comparison

For this subsection assume the two costs are finite convex Young functions, zero at the origin and at least the identity. Write $L = 2^p$, $Q = L^2$, and $h = L^{-1}$. On $G_L = \{1, \dots, L\}^2$ use the modular

$$\mathcal{E}_L(f) = \sum_{\text{horizontal edges}} \theta_1(|\Delta f|) + \sum_{\text{vertical edges}} \theta_2(|\Delta f|).$$

All sums here contain only internal edges; no boundary edges are added. For a continuum function u on $[0,1]^2$, let $f_v = L \operatorname{av}_{C_v} u$, where C_v are the h squares. Jensen along a segment joining adjacent squares gives

$$\mathcal{E}_L(f) \le L^2 \int_{[0,1]^2} \{\theta_1(|u_x|) + \theta_2(|u_y|)\}. \tag{2.1}$$

Indeed each segment average has total overlap at most one after summing over internal edges. If $W(C_v) = hg_v$ with independent standard Gaussians g_v, then, exactly,

$$\sum_v g_v(f_v - \overline{f}) = L^2 W(P_h(u - \overline{u})). \tag{2.2}$$

Thus a continuum histogram lower bound transfers to the grid without an approximation or an extra boundary cost.

For the reverse direction, interpolate hf_v bilinearly between cell centres, and extend constantly in the outward normal direction across the boundary half strips. Call the result u_f. Each derivative is a convex combination of the corresponding grid differences, giving

$$\int (\theta_1(|(u_f)_x|) + \theta_2(|(u_f)_y|)) \le h^2 \mathcal{E}_L(f).$$

The nodal hat functions all have integral h^2, including at the boundary with this extension. Consequently $\overline{u_f} = h\overline{f}$, and the elementary mass-matrix bounds give

$$ch^4 \sum_v (f_v - \overline{f})^2 \le \|u_f - \overline{u_f}\|_2^2 \le Ch^4 \sum_v (f_v - \overline{f})^2.$$

Choose a fixed sufficiently fine dyadic subdivision $\delta = h/M$. The polynomial inverse inequality $\|\nabla u_f\|_2 \le Ch^{-1}\|u_f\|_2$, followed by cell Poincare, shows that $\|P_\delta u_f\|_2 \ge c\|u_f\|_2$ for a fixed large M. The same holds for differences of interpolants and after subtracting constants. Gaussian comparison therefore bounds the grid Gaussian width by CL^2 times the continuum histogram width at mesh δ. Since $\log(1/\delta) = \log L + O(1)$, this preserves the required scale.

2.2 Gaussian width and the empirical process

The following comparisons apply to any symmetric, constant-invariant class of lattice functions with bounded diameter modulo constants, including every class considered here. They require no regularity of the original costs. Assume the law μ on G_L satisfies $\max_v \mu(v) \le A/Q$, that $N \max_v \mu(v) \le Am_0$, and $m_0 \ge c_0 \log(2Q)$. Constants in this subsection may also depend on this fixed positive c_0. Let $\mathcal{H}_L = \{f : \mathcal{E}_L(f) \le Q\}$, and set

$$w_L = \mathbb{E} \sup_{f \in \mathcal{H}_L} \left| \sum_v g_v (f_v - \overline{f}) \right|.$$

Then

$$\mathbb{E} \sup_{f \in \mathcal{H}_L} \left| \sum_{i=1}^{N} (f(U_i) - \mu f) \right| \le C_A \sqrt{m_0}\, w_L. \tag{2.3}$$

To prove this, first replace each f by $f - \mu f$; constants must be removed before symmetrization. Conditional on the occupancy counts N_v, symmetrization and the Rademacher-to-Gaussian comparison give a constant times the Gaussian process with coordinates $\sqrt{N_v}\, g_v$. Its increment metric is bounded by $\sqrt{\max_v N_v}\, \|P_\mu(f - f')\|_2$, where $P_\mu f = f - (\mu f)\mathbf{1}$. On vectors of uniform mean zero,

$$\|P_\mu f\|_2 \le (1 + \sqrt{Q}\|\mu\|_2)\|f\|_2 \le (1 + \sqrt{A})\|f\|_2.$$

Gaussian comparison now bounds its width by $C_A \sqrt{\max_v N_v}\, w_L$. Finally the exponential moment bound for a binomial count and a union bound imply $\mathbb{E} \max_v N_v \le C_A(m_0 + \log(2Q)) \le C_{A,c_0} m_0$. Cauchy–Schwarz proves (2.3).

There is also a converse for the uniform law, valid already for $N \ge Q$:

$$\mathbb{E} \sup_{f \in \mathcal{H}_L} \left| \sum_{i=1}^{N} (f(U_i) - \overline{f}) \right| \ge c \sqrt{N/Q}\, w_L. \tag{2.4}$$

Here are details, since this avoids any limit in the sampling density. For $\lambda \ge 1$, let $X = P - P'$ with independent Poisson variables of mean λ. There are universal constants $c, C > 0$ such that

$$\mathbb{P}(X \ge t) \ge c \exp(-Ct^2/\lambda), \qquad t \ge \sqrt{\lambda}.$$

Indeed $\mathbb{P}(P' \le \lambda) \ge c$. Stirling's formula, summed over $\lfloor \sqrt{\lambda} \rfloor$ consecutive integers beginning at $\lceil \lambda + t \rceil$, gives the claimed lower bound for $\mathbb{P}(P \ge \lambda + t)$. The prefactor is bounded below by $c(1 + t/\lambda)^{-1/2}$ and is absorbed in the exponential since $t \ge \sqrt{\lambda} \ge 1$. Consequently, for $t \ge 0$,

$$\mathbb{E}(X - t)_+ \ge c\sqrt{\lambda} \exp[-C(t/\sqrt{\lambda} + 1)^2].$$

For a sufficiently small universal $a > 0$, this dominates

$$\mathbb{E}(a\sqrt{\lambda} G - t)_+ \le \frac{a\sqrt{\lambda}}{\sqrt{2\pi}} e^{-t^2/(2a^2\lambda)}.$$

Symmetry gives the same stop-loss comparison for negative t. Thus X dominates $a\sqrt{\lambda} G$ in convex order. Independence and coordinate-by-coordinate replacement extend this comparison to any convex function of finitely many coordinates, in particular the support function of the centered class $\mathcal{H}_L$.

Put $\lambda = N/Q$ and Poissonize the number of observations to $M \sim \mathrm{Pois}(N)$. The cell counts P_v are independent $\mathrm{Pois}(\lambda)$. On uniform-mean-zero tests, the Poissonized support function is the same for (P_v) and $(P_v - \lambda)$. Its independent difference has coordinates distributed as X, so the triangle inequality and the convex-order comparison give

$$2\,\mathbb{E}\Big\| \sum_{i=1}^{M} (\delta_{U_i} - \mu) \Big\|_{\mathcal{H}_L} \geq c\sqrt{\lambda}\, w_L.$$

If $F(k)$ denotes the expected support function with exactly k observations, centered independent increments and Jensen imply that F is nondecreasing, while the triangle inequality implies $F(k+\ell) \leq F(k) + F(\ell)$. Hence $F(k) \leq \lceil k/N \rceil F(N)$ and $\mathbb{E}F(M) \leq 2F(N)$. This proves (2.4). In conjunction with (2.3), the Gaussian and empirical criteria are therefore equivalent in Talagrand's regime, without requiring $m_0 \to \infty$ on a fixed grid.

3 Local geometric and Gaussian constructions

This section proves the local facts used by the adaptive flow construction. The geometry is described in a determinant-one chart. Thus chart area is physical area. A parent is a parallelogram of diameter D containing a ball of radius $c_* D$, with fixed $c_* > 0$. Terminal pieces are not refined. All constants below may depend on c_* and on fixed contraction and Hölder exponents, but not on the mesh or the chart eccentricity in physical coordinates.

Lemma 3.1 (Gaussian concentration)**.** *If F is an L-Lipschitz function of a standard Gaussian vector, then*

$$\mathbb{E}e^{t(F-\mathbb{E}F)} \leq e^{t^2 L^2/2}, \qquad \mathbb{P}\{F > \mathbb{E}F + u\} \leq e^{-u^2/(2L^2)}.$$

The same holds for a supremum of Gaussian linear tests whose test vectors have norm at most L, whenever its expectation is finite.

Proof. For a smooth positive f and the Gaussian semigroup $P_t f(x) = \mathbb{E}f(e^{-t}x + \sqrt{1-e^{-2t}}Z)$, differentiation and Gaussian integration by parts give

$$\mathrm{Ent}_\gamma(f) = \int_0^\infty \mathbb{E}\frac{|\nabla P_t f|^2}{P_t f}\, dt.$$

One obtains the identity first for bounded positive smooth functions by differentiating $\mathbb{E}[(P_t f)\log(P_t f)]$; its limit at infinity is $(\mathbb{E}f)\log(\mathbb{E}f)$. Since $\nabla P_t f = e^{-t} P_t \nabla f$, Cauchy–Schwarz under P_t gives

$$\mathrm{Ent}_\gamma(f) \leq \frac{1}{2}\mathbb{E}\frac{|\nabla f|^2}{f}.$$

Apply this to $f = e^{tF}$ and use $|\nabla F| \leq L$. If $A(t) = \log \mathbb{E}e^{tF}$, it follows that $tA'(t) - A(t) \leq t^2 L^2/2$. Integrating the derivative of $A(t)/t$, whose limit at zero is $\mathbb{E}F$, proves the moment bound. Exponential Markov gives the tail. Smooth approximation and truncation extend the result to Lipschitz functions. Finite maxima of linear tests are Lipschitz with the stated constant; increasing finite approximations and then truncation prove the last assertion. □

Lemma 3.2 (A subdivision with a minimum side)**.** *Fix a sufficiently small constant $\tau > 0$ and a minimum side $s > 0$. If $\tau D < Cs$, the parent may be declared terminal. Otherwise it has a*

measurable subdivision into square children and terminal children. Give a square its side l_S, and a terminal child the label $l_S = s$. Then

$$|S| \asymp l_S^2, \quad \operatorname{diam} S \lesssim l_S, \quad l_S \asymp \max\{s, \min(\tau D, d_Q(x))\} \quad (x \in S), \qquad d_Q(x) = \operatorname{dist}(x, \partial Q).$$

For a sufficiently small fixed $c > 0$, the neighborhoods $S^+ = \{x \in Q : \operatorname{dist}(x, S) < cl_S\}$ have bounded overlap; intersecting neighborhoods have comparable labels. If $l < l'$ are labels occurring in one parent, the children of label l lie in a boundary strip of width Cl. A ball of radius Cl' therefore meets at most $C'l'/l$ such children. At a fixed larger label, only $O(1)$ corresponding neighborhoods can meet a given smaller neighborhood.

The square children contract by an arbitrarily small fixed factor when τ is chosen small. After any chart change whose relative metric eigenvalues lie in a fixed compact interval, the same construction has uniform constants.

Proof. Use a dyadic lattice of smallest side comparable to s; changing s by a factor two is harmless. Select maximal lattice squares of side $2s \le l \le \tau D$ contained in Q and satisfying $\operatorname{dist}(S, \partial Q) \ge \Gamma l$, where Γ is a large fixed constant. Their interiors are disjoint. Every remaining piece lies in a smallest lattice square, either a full square in Q or a clipped one. A full uncovered square has distance $O(s)$ from the boundary: its $2s$ ancestor is under the upper cap and otherwise would satisfy both containment and clearance.

There is a full uncovered smallest square within $O(s)$ of every clipped one. To see this, choose a point x in the clipped piece and move a fraction As/D toward an inball center. Convexity supplies a ball of radius Ac_*s at the resulting point. Choose A large enough to contain its lattice square and choose the stopping constant large enough that the fraction is less than one. The resulting full square is still within $O(s)$ of the boundary. Taking Γ larger than this fixed constant ensures it is uncovered. Assign each clipped piece to one such full square, breaking ties in a fixed lattice order. Only $O(1)$ pieces can be assigned to one receiver, because their lattice squares lie within $O(s)$ of it. The resulting terminal children have area comparable to s^2 and diameter $O(s)$. They partition the uncovered region.

For a selected square, its dyadic parent fails containment or clearance, or exceeds the upper cap. In the first two cases $d_Q \lesssim l_S$ throughout the square; in the last $l_S > \tau D/2$. Clearance gives the converse bound. Together with the terminal estimates this proves the displayed comparison. The function $x \mapsto \max\{s, \min(\tau D, d_Q(x))\}$ is 1-Lipschitz. Choosing c small in terms of the comparison constants shows that intersecting neighborhoods have comparable labels. Disjoint child areas and packing then give bounded overlap.

If a smaller label $l < l'$ occurs, the parent of an l square cannot fail the upper cap, since $2l \le l' \le \tau D$. It must consequently lie in a boundary strip of width Cl; the same is true of terminal children. Inside a ball of radius Cl', the four straight boundary strips have total area $O(ll')$. Dividing by child area $\asymp l^2$ proves the $O(l'/l)$ count. For a fixed larger label, disjoint areas of size $\asymp (l')^2$ give the $O(1)$ reverse count. Fixed enlargements only change the constants. All tests, lattice choices and tie rules are measurable. A bounded relative chart change preserves an inball and diameter comparison, proving the final assertion. □

Lemma 3.3 (Predictable bridges and a local ray inverse)**.** *Choose each subdivision from the previously revealed Gaussian masses. Conditionally on this history, the new child-average contrast has its orthogonal Gaussian law and the remaining noises are independent mean-zero white-noise bridges in the children.*

For such a subdivision there is a continuous zero-normal current V whose divergence is the child-average contrast. On every S^+ there is an even envelope E_S controlling

$$\sup_{S^+} |V| + l_S^{\zeta}[V]_{C^{\zeta}(S^+)} + l_S \| \operatorname{div} V \|_{L^{\infty}(S^+)}, \qquad 0 < \zeta < 1,$$

with

$$\mathbb{E}(E_S \mid \text{old}) \le C, \qquad \mathbb{P}(E_S > t \mid \text{old}) \le C e^{-ct^2}.$$

The pointwise chart covariance of V is at most CI. Each of the linear tests defining the envelope, before subtracting the parent constant, has a kernel bounded by C/l_S and supported within distance Cl_S of S. The envelope is a fixed sum of suprema of such tests.

Proof. The bridge assertion is orthogonal Gaussian projection. Given the old history, the child indicators are fixed tests in the old parent bridge. Their centered span and its orthogonal complement are independent Gaussian subspaces; the complement is the direct sum of child bridges. Induction proves the assertion for predictable choices. One may construct the conditional laws on a countable dense collection of tests and extend by mean-square continuity; measurability follows from the measurable indicator maps of the subdivisions.

Let η_Q be a smooth probability density in a central ball of Q, with $\|\eta_Q\|_{\infty} \lesssim D^{-2}$ and $\|\nabla \eta_Q\|_{\infty} \lesssim D^{-3}$. Set

$$K_Q(x,y) = (x-y) \int_1^{\infty} t \eta_Q(y + t(x-y))\, dt.$$

Transport a unit source at y along its straight segment to a point z averaged with density $\eta_Q(z)$. Its divergence is $\delta_y - \eta_Q$, and changing variables along the segment gives this kernel. Convexity keeps the segment in Q; extension by zero has no outer normal flux. Direct differentiation, using $t|x-y| \le CD$ wherever the integrand is nonzero, gives

$$|K_Q(x,y)| \lesssim |x-y|^{-1}, \qquad |\nabla_x K_Q(x,y)| + |\nabla_y K_Q(x,y)| \lesssim |x-y|^{-2}.$$

There is also the useful support property

$$K_Q(x,y) = 0 \quad \text{if } |x-y| > C d_Q(x).$$

Indeed $x = (1-t^{-1})y + t^{-1}z$, where z is in the central ball. A nearest supporting line to x has distance at least cD from z and nonnegative distance from y, so $d_Q(x) \ge cD/t$, while $|x-y| \le CD/t$. In particular $\int |K_Q(x,y)| dy \lesssim d_Q(x)$, which shows that the inverse of a bounded source is continuous and vanishes on the boundary. More explicitly, splitting into disks of radius $2a$ about two poles and the remaining annulus yields

$$\int |K_Q(x,y) - K_Q(x',y)| dy \lesssim a(1 + \log(R/a)), \quad a = |x-x'| \le R/2,$$

when both supports lie in a ball of radius CR.

Let P be child averaging and P_Q parent averaging. The contrast current is $V(x) = W((P - P_Q)K_Q(x,\cdot))$. Its divergence is the required contrast. For $x, x' \in S^+$, take $R = l_S$. A child meeting a sufficiently small ball about x has side comparable to l_S by Lemma 3.2; averaging the $|x-y|^{-1}$ singularity there gives C/l_S. Farther children satisfy the same bound directly. Every child meeting the kernel support has diameter $O(l_S)$, so averaging enlarges that support by only $O(l_S)$. The

two-pole estimate, divided by the near child area, and the derivative bound on far children give, for any $\beta < 1$,

$$\|PK_Q(x,\cdot)\|_\infty \le C/l_S, \quad \left\| \left(\frac{l_S}{|x-x'|} \right)^\beta P(K_Q(x,\cdot) - K_Q(x',\cdot)) \right\|_\infty \le C_\beta / l_S.$$

These kernels have support area $O(l_S^2)$ and therefore bounded L^2 norms. The normalized density test is $l_S P(x,\cdot)$; it has exactly the same bounds. Subtracting P_Q is a common orthogonal projection and contracts L^2 norms. This proves the covariance assertion.

Choose $\zeta < \beta < 1$. On rescaling S^+ to diameter $O(1)$, nets of mesh 2^{-j} have cardinality $O(2^{2j})$. The maximum of the Gaussian differences between successive nets and neighboring net points has expectation $O(2^{-\beta j}\sqrt{1+j})$. Summing after multiplication by $2^{\zeta j}$ gives finite expected supremum and C^ζ seminorm. Only $O(1)$ comparable children meet S^+, so its normalized density maximum has bounded expectation as well. Each envelope test has bounded L^2 norm; Lemma 3.1 gives the tail. Taking absolute suprema makes the envelope even in all fresh Gaussian coordinates. The localized preprojection kernel bounds just proved apply to every test in these suprema. □

Lemma 3.4 (Joint envelope tails)**.** *Conditionally on the old history, for every predictable finite set $\mathcal{A}$ of attempted children and all sufficiently large t,*

$$\mathbb{P}\{E_S > t \text{ for every } S \in \mathcal{A} \mid \text{old}\} \le e^{-c|\mathcal{A}|t^2}.$$

Proof. Choose one signed test kernel k_S per child, before subtracting parent constants. If their supports overlap and $l = l_S \le l_T = l'$, then

$$|\langle k_S, k_T \rangle| \le Cl/l'.$$

For a fixed larger child, Lemma 3.2 bounds the number of overlapping smaller tests at side l by Cl'/l. For a fixed smaller child, only $O(1)$ tests at each larger side can overlap. Apply the weighted Schur test with weights l_S^η, $0 < \eta < 1$. For a row at the larger scale, the contribution from scale l is at most $C(l/l')^\eta$; for a row at the smaller scale, the contribution from l' is at most $C(l/l')^{1-\eta}$. Both geometric sums are bounded. Comparable scales are controlled by packing. Thus

$$\Big\| \sum_S c_S k_S \Big\|_2^2 \le C \sum_S |c_S|^2$$

uniformly over all choices of tests. Within each parent, subtracting its constant is a common orthogonal projection and cannot increase this norm. Different parents are disjoint Gaussian source blocks, so the same inequality holds across all parents and their different charts.

The sum of envelopes is a fixed sum of suprema over choices of one test per child. The preceding bound makes it $C\sqrt{|\mathcal{A}|}$- Lipschitz in the fresh Gaussian vector. Its conditional mean is $O(|\mathcal{A}|)$. The event in the lemma forces this sum above $t|\mathcal{A}|$. Gaussian concentration proves the claim. □

Lemma 3.5 (Even clipping and exact omitted sources)**.** *Fix a threshold E_0. The fresh current can be multiplied pointwise by an even scalar in $[0,1]$ so that every retained childwise current and scaled Hölder envelope is at most CE_0. Its omitted divergence is a sum of bounded mean-zero sources f_S, one per exceptional child, with*

$$\|f_S\|_\infty \le C\sigma E_S / l_S, \qquad \|f_S\|_1 \le C\sigma E_S l_S.$$

Here σ is the scale multiplying the fresh current, and f_S is supported in the physical image of an intrinsic ball of radius Cl_S intersected with its parent. Its associated omitted current has zero outer normal trace, and in particular the source has integral zero. At a fixed point this attenuation does not increase the exponential moment of any fixed linear functional of the fresh Gaussian current.

Proof. Call S exceptional when $E_S > E_0$. Take a Lipschitz cutoff χ_S equal to one on S, supported in S^+, with chart gradient at most C/l_S. Multiply V by $a = \prod_{S \text{ exceptional}}(1-\chi_S)$. The factor is even and in $[0,1]$. It vanishes on exceptional children. On a nonexceptional child, only boundedly many cutoffs occur, at comparable sides. The product difference inequality and their Lipschitz bounds show that the retained supremum and scaled Hölder seminorm are at most $CE_S \le CE_0$.

Order the exceptional children and write $1-a = \sum_S \varphi_S$, where $\varphi_S = \chi_S \prod_{T<S}(1-\chi_T)$. Bounded comparable overlap gives $|\nabla\varphi_S| \le C/l_S$. Define $f_S = \sigma \operatorname{div}(\varphi_S V)$. The product rule and the envelope's density bound give the asserted L^∞ bound; the support has area $O(l_S^2)$, giving the L^1 bound. The uncut ray current is continuous and vanishes on the parent boundary. Therefore the product's zero extension has no boundary measure in its divergence, and its integral is zero. Determinant-one pushforward preserves these statements.

For the moment assertion let g be the entire fresh Gaussian vector and $\ell(g)$ the fixed linear functional at the point. Since $a(-g) = a(g)$,

$$\mathbb{E}e^{a(g)\ell(g)} = \mathbb{E}\cosh(a(g)\ell(g)) \le \mathbb{E}\cosh(\ell(g)) = \mathbb{E}e^{\ell(g)}.$$

□

Lemma 3.6 (Sparse exact repair)**.** *Let $m = \log(1/h)$. Suppose there are at most Cm generations, every chart and its inverse have norm at most m^C, and intrinsic child labels lie in $[h, m^C]$. The domain is the unit square; the same proof applies to an area-one axis rectangle with polynomially bounded sides and reciprocals. Assume*

$$\log m = o(E_0^2), \qquad E_0^2 \le m, \quad D_0 = m/E_0^2, \qquad T_0 = \sigma\sqrt{m}.$$

For every $1 \le B \le m^C$, with probability $1-o(1)$ all omitted sources in Lemma 3.5 have a combined exact zero-normal repair supported on a set of area at most $e^{-cE_0^2}$, and

$$\|J_y\|_\infty \le CT_0D_0/B, \qquad \|J_x\|_{\psi_2} \le CT_0D_0B/E_0, \qquad \psi_2(t) = e^{t^2}-1.$$

The conclusion remains valid before any stopping rule based on the previously revealed history.

Proof. Each child has area at least ch^2, so there are at most Cmh^{-2} attempts, up to polynomial factors. The conditional Gaussian tails and a union bound show $E_S \le C\sqrt{m}$ for every attempt, with probability $1-o(1)$. On this event an omitted source in a determinant-one ellipse of intrinsic radius Cl satisfies

$$\|f_S\|_\infty \le CT_0/l, \qquad \|f_S\|_1 \le CT_0 l.$$

Write $G = AA^{\mathsf{T}}$ for its chart metric. The maximal horizontal and vertical chords of this ellipse are bounded by $Cl/\sqrt{G_{yy}}$ and $Cl/\sqrt{G_{xx}}$. This follows by setting the other coordinate equal to zero in its quadratic form, since $\det G = 1$. Thus the maximal absolute horizontal and vertical line masses, A_x, A_y, satisfy $A_xA_y \le CT_0^2$.

Choose, before the fresh reveal, an axis box in the domain containing the support ellipse portion, with both sides at least Bl. Enclose its coordinate projections, enlarge, and slide the intervals inside the domain. Each side is at most a polynomial times l. If such a box cannot fit, then $l \ge m^{-C'}$;

area packing leaves only polynomially many such candidates per generation. The probability any of them is exceptional is $o(1)$, since $E_0^2 \gg \log m$.

In a box $I \times J$ and for any bounded mean-zero f, set

$$r(y) = \int_I f(x,y)dx, \qquad J_x(x,y) = \int_{\inf I}^{x} \left(f(t,y) - \frac{r(y)}{|I|} \right) dt, \quad J_y(x,y) = \frac{1}{|I|} \int_{\inf J}^{y} r(t)dt.$$

These primitives have divergence f and zero normal boundary trace. If $A_x \le T_0 B$, this gives horizontal supremum $2A_x \le CT_0B$ and vertical supremum at most $\|f\|_1/|I| \le CT_0/B$. If $A_x > T_0B$, then $A_y \le CT_0/B$ and the column version gives the same two bounds. Extending each box current by zero preserves its divergence.

We now control overlap, including dependence across generations. Fix a point and pad all predictable boxes by $O(h^2)$. In one generation at most $m^{C'}$ candidate boxes contain the point: at each intrinsic dyadic size their disjoint child areas lie within a polynomial times that size of the point, and there are $O(m)$ size classes. Let I_S indicate exceptionality among these candidates. For $\lambda = cE_0^2/2$, the joint tail lemma and expansion over subsets give

$$\mathbb{E}\left(e^{\lambda \sum_S I_S} \mid \text{old}\right) = \mathbb{E}\left(\prod_S [1 + (e^\lambda - 1)I_S] \mid \text{old}\right) \le \exp\{m^{C'} e^{-cE_0^2/2}\}.$$

Iterate over the $O(m)$ generations, adding empty ones after a stopping rule. Chernoff's inequality bounds the probability of overlap exceeding $C'D_0$ by $\exp(-c'C'm + o(1))$. A union bound over a mesh of spacing h^2 and the padding gives a uniform overlap bound CD_0 for a sufficiently large fixed C'. Padding is harmless in the candidate count because physical sizes are at least h times an inverse polynomial, which exceeds h^2.

The sum of candidate box areas in one generation is polynomially bounded: each is at most a polynomial times l_S^2, while the child areas are disjoint. Conditional single-child tails therefore bound the expected total exceptional-box area over all generations by $m^{C''} e^{-cE_0^2}$. Markov's inequality makes this area at most $e^{-c'E_0^2}$ with probability $1 - o(1)$. The uniform overlap gives hard supremum CT_0D_0/B and other supremum $M = CT_0D_0B$ on this sparse support. A function bounded by M on area q has

$$\|f\|_{\psi_2} \le \frac{M}{\sqrt{\log(1 + 1/q)}}.$$

Substitution in its Luxemburg modular proves this inequality and the claimed soft bound. Every repair is exact and zero-normal, so their sum has the stated divergence. □

Corollary 3.7 (Sparse repair in two Orlicz norms)**.** *In the setting of Lemma 3.6, let ψ_i be Young functions with generalized inverses ρ_i. Suppose*

$$\rho_1(e^u)\rho_2(e^u) \ge c\sqrt{u} \quad \textit{for all sufficiently large } u.$$

Then, for $E_0 = m^{1/2-\eta}$ and $0 < \eta < 1/10$, the omitted normalized sources, with $T_0 = O(1)$, have an exact repair satisfying

$$\max_i \|J_i\|_{L^{\psi_i}} \le Cm^{(5/2)\eta - 1/4} = o(1)$$

with probability $1 - o(1)$.

Proof. The proof of the sparse-repair lemma gives, on area at most $q = e^{-c_0 E_0^2}$, the two supremum bounds CT_0D_0B and CT_0D_0/B. It is symmetric under exchanging axes, so the same statement holds for polynomially bounded positive B and B^{-1}. Put

$$R_i = \min\{\rho_i(e^{c_0 E_0^2}), m\}, \qquad B = \sqrt{R_1/R_2}.$$

Each R_i is bounded below by a fixed positive constant for large m. If neither truncation is active their product is at least $c'E_0$. If a truncation is active, their product is at least $c'm \geq c'E_0$. Thus B, B^{-1} are polynomially bounded and $R_1R_2 \geq c'E_0$. A function bounded by M on area q has Luxemburg norm at most $M/\rho_i(1/q)$. Consequently both repair norms are at most

$$\frac{CT_0D_0}{\sqrt{R_1R_2}} \leq \frac{Cm}{E_0^{5/2}} = Cm^{(5/2)\eta - 1/4}.$$

□

Lemma 3.8 (Grid capacity and Gaussian repair)**.** *On the graph of a square of K^2 cells, with K a power of two, orient internal faces and let dq be outgoing minus incoming load at each cell. For $\sum_v g_v = 0$,*

$$\mathcal{B}(g) := \min_{dq=g} \max_e |q_e| = \max_{\varnothing \neq S \neq D} \frac{|\sum_{v\in S} g_v|}{|\partial S|} \leq C\|g\|_2.$$

Consequently $\mathcal{B}$ is C-Lipschitz. If g is centered Gaussian with covariance at most the projection onto mean-zero vectors, then

$$\mathbb{E}\mathcal{B}(g) \leq C(1+\log K), \qquad \mathbb{P}\{\mathcal{B}(g) > C(1+\log K) + t\} \leq e^{-ct^2}.$$

For physical cell width w, interpolating a component linearly between its two normal face loads gives a zero-normal continuum current with component suprema at most $\mathcal{B}(g)$ and divergence g_v/w on cell v.

Proof. Any feasible flow gives the cut lower bound by summing divergence over S. Conversely, the image under d of the face cube $[-b,b]^E$ has support function $b\sum_{v\sim z}|u_v - u_z|$ at a vertex function u. Subtract $\min u$ and integrate superlevel sets. The cut inequalities imply $\langle g, u\rangle \leq b\sum_{v\sim z}|u_v - u_z|$ for every u; finite-dimensional convex separation therefore puts g in that image. This proves equality.

For $|S| \leq K^2/2$, if no occupied row or column is full, each occupied row and column is mixed. Their numbers have product at least $|S|$, so their cut edges give $|\partial S| \geq c\sqrt{|S|}$. If a row is full, all columns are occupied and at least $K/2$ are mixed, since at most $|S|/K \leq K/2$ columns can be full. The same argument handles a full column. Thus in all cases $|\partial S| \geq c\sqrt{\min(|S|, |D\setminus S|)}$. Cauchy–Schwarz in the cut formula proves the Euclidean bound, and subadditivity gives its Lipschitz assertion.

For independent standard Gaussians Z_v, decompose their centered vector into orthogonal dyadic child-average contrasts. In a parent of side L cells, each of its four child averages has standard deviation $O(1/L)$. Subtract row means and take horizontal primitives, then take vertical primitives of those row means. The result routes the parent contrast with zero outer load. Each face load in this one-level flow has variance $O(1)$, because its coefficients on child values are $O(L)$. Different levels are orthogonal Gaussian subspaces, so the summed flow has face variance $O(1+\log K)$. There are $O(K^2)$ faces; the Gaussian maximum estimate gives expected supremum $C(1+\log K)$. For covariance dominated by the mean-zero projection, add an independent Gaussian with the covariance difference and use convexity of $\mathcal{B}$ to obtain the same expected bound. Gaussian concentration and its Euclidean Lipschitz constant prove the tail. Linear normal-face interpolation has matching traces across internal faces, zero exterior normal trace, and divergence equal to the face-load difference divided by w, proving the last statement. □

Lemma 3.9 (Collected local tools)**.** *The ordinary adaptive refinement may use a subdivision with any fixed sufficiently small intrinsic contraction, a pointwise chart covariance bound, even child*

envelopes with uniform Gaussian and joint tails, even clipping with exact omitted sources, the sparse-repair bounds of Lemma 3.6, and the dimension-independent grid capacity and Gaussian repair bounds of Lemma 3.8. All remain valid conditionally on previous reveals and before predictable stopping rules.

Proof. This is precisely the content of Lemmas 3.2– 3.8. Their proofs use only conditional orthogonal projection, the parent geometry, and deterministic kernel and packing bounds, so their constants do not depend on the realized past or the physical eccentricities allowed in the sparse-repair lemma. □

4 Constructing the two-cost phase

4.1 Regularizing arbitrary Young widths

No regular growth assumption on the Young functions is needed. The following reduction supplies it while shrinking the permissible widths. For squared widths $b_{i,j} = \rho_i(e^{2^j})^2$ put

$$v_{i,j} = \frac{2^{j/2}}{\sqrt{b_{i,j}}}, \qquad G_{i,J} = \mathbb{E}\max_{j\le J} v_{i,j}|g_j|.$$

The two prefix entropy conditions are equivalent, up to their fixed constants, to

$$G_{1,J} \le C\sqrt{b_{2,J}}, \qquad G_{2,J} \le C\sqrt{b_{1,J}} \qquad (J \ge 1). \tag{4.1}$$

First the full family of prefix estimates implies $\inf_j p_j > 0$, as shown in the first paragraph of the allocation lemma below. Thus $v_{i,j}/\sqrt{b_{3-i,J}} \le C$ for $j \le J$, by monotonicity. Under this largest-coordinate bound, the standard Gaussian-maximum equivalence applies: if $\sum_j \exp(-cR^2/v_j^2) \le C$ and $\max_j v_j \le C_0 R$, a union bound integrated above a sufficiently large multiple of R gives $\mathbb{E}\max_j v_j|g_j| \le C'R$. Conversely, if this expectation is at most CR, the indices with $v_j \ge t$ have cardinality at most $\exp(C'R^2/t^2)$, because the expected maximum of M independent standard Gaussian absolute values is at least $c\sqrt{\log(M+1)}$. Partitioning the v_j into dyadic bands gives the exponential sum bound with a sufficiently large exponent constant. These arguments apply uniformly to every finite prefix. The harmless largest-coordinate issue can alternatively be read directly from $G_{i,J} \ge \sqrt{2/\pi}\, v_{i,j}$.

Fix $q > 1$. Regularize one squared width by

$$\widetilde{b}_{1,j} = \inf_{k\le j} 2^{q(j-k)} b_{1,k}. \tag{4.2}$$

It is nondecreasing, satisfies $1 \le \widetilde{b}_{1,j} \le b_{1,j}$, and $\widetilde{b}_{1,j+1} \le 2^q \widetilde{b}_{1,j}$. We claim that both inequalities in (4.1) remain valid with this new width. Write $r = 2^{-(q-1)/2} < 1$. Its new reciprocal width is

$$\widetilde{v}_{1,j} = \sup_{k\le j} r^{j-k} v_{1,k}.$$

Consequently

$$\begin{aligned}\mathbb{E}\max_{j\le J} \widetilde{v}_{1,j}|g_j| &\le \sum_{\ell\ge 0} r^\ell \mathbb{E}\max_{k\le J-\ell} v_{1,k}|g_{k+\ell}| \\ &\le (1-r)^{-1} G_{1,J} \le C\sqrt{b_{2,J}}.\end{aligned}$$

For the other inequality, take an index $k \le J$ achieving $\widetilde{b}_{1,J} = 2^{q(J-k)} b_{1,k}$. The old prefix inequalities imply the pointwise product lower bound $b_{1,k} b_{2,k} \ge c 2^k$. For $\ell > k$, monotonicity of b_2 gives

$$v_{2,\ell} \le 2^{(\ell-k)/2} v_{2,k} \le C 2^{(\ell-k)/2} \sqrt{b_{1,k}}.$$

The expected maximum over these later indices is therefore at most $C 2^{(J-k)/2} \sqrt{b_{1,k}}$, using a geometric sum. The earlier indices contribute $G_{2,k} \le C\sqrt{b_{1,k}}$. Since $q > 1$,

$$G_{2,J} \le C 2^{q(J-k)/2} \sqrt{b_{1,k}} = C\sqrt{\widetilde{b}_{1,J}}.$$

This proves the claim. Applying the same operation to the second width preserves both inequalities once again.

Interpolate the logarithms of the resulting widths linearly at $s_j = j \log 2$. They are nondecreasing and have Lipschitz constant at most q. If desired, convolve these logarithms with one fixed smooth positive kernel. This changes each width by at most a fixed factor, preserving the prefix conditions. At every intermediate s, the resulting width is bounded by a fixed multiple of the original $\rho_i(e^{e^s})^2$: use the left endpoint of the containing interval, monotonicity of the original width, and the factor 2^q between the two new endpoints. Thus the regularized widths may be used for an upper bound for the original Young functions.

4.2 From prefix entropy to smooth squared semiaxes

The following elementary construction is useful because it permits the preferred coordinate direction to change arbitrarily often. Write $s_j = j \log 2$ and

$$b_{i,j} = \rho_i(e^{2^j})^2, \qquad p_j = b_{1,j} b_{2,j} 2^{-j}.$$

Assume the two prefix estimates

$$\sup_J \sum_{j \le J} \exp[-c\, b_{1,j} b_{2,J} 2^{-j}] < \infty, \qquad \sup_J \sum_{j \le J} \exp[-c\, b_{2,j} b_{1,J} 2^{-j}] < \infty \tag{4.3}$$

for some $c > 0$. Increasing c does not invalidate these estimates. Suppose also that the widths have been regularized in logarithmic scale, so that there are positive nondecreasing functions $b_i(s)$, dominated by the original Young widths up to fixed factors, satisfying the same prefix estimates at s_j, and for which $\log b_i$ are Lipschitz. All conclusions below allow fixed factors in the upper bounds on the widths.

Lemma 4.1 (Smooth allocation of the two widths)**.** *For every sufficiently small $\eta > 0$ there are smooth positive functions $a(s), b(s), T(s)$ on a half line $s \ge s_0$, and constants $c_*, \varepsilon_*, C_*, K > 0$, such that*

$$a(s) \le C_* b_1(s), \qquad b(s) \le C_* b_2(s), \qquad a(s) b(s) = e^s T(s), \tag{4.4}$$

$$K \le T(s) \le C_* \log(e + s - s_0), \qquad |T^{(q)}(s)| \le C_q \eta \quad (q \ge 1), \tag{4.5}$$

$$\alpha = (\log a)', \quad \beta = (\log b)', \qquad \alpha, \beta \ge \varepsilon_* e^{-c_* T}, \qquad \alpha + \beta = 1 + T'/T, \tag{4.6}$$

$$|\alpha^{(q)}| \le C_q \alpha, \qquad |\beta^{(q)}| \le C_q \beta \quad (q \ge 1). \tag{4.7}$$

Here K can be prescribed beforehand. The constants may depend on the prefix bounds, the width regularization, and the chosen parameters. The coefficient of η in the derivative bounds is fixed; by starting with a smaller parameter it can be made as small as subsequently needed.

Proof. First, $\inf_j p_j > 0$. Indeed, monotonicity of the widths gives, for $j - r \le k \le j$,

$$b_{1,k} b_{2,j} 2^{-k} \le 2^r p_j.$$

If p_j tends to zero, the number of preceding indices for which this quantity is at most one tends to infinity, contradicting the first prefix estimate. There are enough preceding indices because the widths are bounded below by a positive constant. Hence the continuous function $p(s) = b_1(s) b_2(s) e^{-s}$ is bounded below as well, by local comparability.

For each $i = 1, 2$, the prefix estimates imply the interval bound

$$\sum_{r \le j \le J} e^{-c_0 p_j} \le C \left[1 + \log \frac{b_{i,J}}{b_{i,r}} \right]. \tag{4.8}$$

To prove this, partition the indices by the dyadic bands of $b_{i,j}$. Within any one band let J' be its last index. The appropriate prefix estimate at J' bounds the sum over this band of e^{-2cp_j}, since $b_{i,J'}/b_{i,j} \le 2$. The number of bands is at most a constant times $1 + \log(b_{i,J}/b_{i,r})$. This proves (4.8) with $c_0 = 2c$. Equivalently, the same argument gives the two weighted Carleson bounds

$$\sum_{j \le J} \frac{b_{i,j}}{b_{i,J}} e^{-c_0 p_j} \le C.$$

The constants can be chosen without a circular dependence on the desired small slope η. First choose the fixed exponential clock coefficient. Then take $K = K(\eta)$ so large that the factors $e^{-c_1 K}/\eta^r$ occurring below, for the finitely many required powers r, are bounded by fixed constants. The exponential-kernel boundary sums and the additional caps then have uniform interval-clock constants. In particular the small multiplier ε below can be chosen independently of the subsequently needed smallness of η. Increasing A ensures the prescribed lower bound K, and only changes fixed width and Hall padding constants; such constants are permitted here.

Choose a large constant A and define the Lipschitz lower envelope

$$T_0(s) = \inf_{j \ge 1} \{ A p_j + \eta |s - s_j| \}.$$

By increasing A, its lower bound is at least the prescribed K; local comparability gives $T_0(s) \le C p(s)$. Its Lipschitz constant is at most η. Choose constants A_1, R with $A_1 c_1 > 2$ and $R \ge A_1/\eta$, where $c_1 > 0$ is fixed, and put

$$T_{\mathrm{cap}}(s) = K + A_1 \log(1 + (s - s_0)/R), \qquad T = \min\{T_0, T_{\mathrm{cap}}\}.$$

Adjusting s_0 by a fixed amount, if necessary, leaves all estimates unchanged. This cap is also η-Lipschitz and $\int_{s_0}^{\infty} e^{-c_1 T_{\mathrm{cap}}} < \infty$. Choose A so that $c_1 A \ge c_0$. Since

$$e^{-c_1 T_0(s)} \le \sum_j e^{-c_1 A p_j} e^{-c_1 \eta |s - s_j|},$$

integration over an interval, followed by (4.8), yields for both i

$$\int_v^w e^{-c_1 T(s)}\, ds \le C \left[1 + \log \frac{b_i(w)}{b_i(v)} \right], \qquad w \ge v. \tag{4.9}$$

The indices outside the interval contribute only a fixed geometric boundary sum. The logarithmic Lipschitz bound on the widths absorbs the fixed enlargement of its endpoints. The cap contributes a finite global constant.

Set $m(s) = \varepsilon e^{-c_1 T(s)}$, with $\varepsilon > 0$ sufficiently small, and $\tau(s) = s + \log T(s)$. Then $\tau' \geq 9/10$, $2m < \tau'$, and, for both i,

$$\int_v^w m \leq C_0 + \tfrac{1}{2} \log \frac{b_i(w)}{b_i(v)}. \tag{4.10}$$

Because $T(v) \leq Cp(v)$, a fixed $D > 0$ consequently ensures the two Hall inequalities

$$\tau(v) + \int_v^w m \leq \log b_1(v) + \log b_2(w) + 2D, \qquad \tau(v) + \int_v^w m \leq \log b_2(v) + \log b_1(w) + 2D \tag{4.11}$$

for every $v \leq w$.

Here is an explicit bounded-slope selection, avoiding a compactness argument. Define

$$M(s) = \int_{s_0}^s m, \quad N(s) = \tau(s) - \tau(s_0) - 2M(s), \quad L(s) = \tau(s) - \log b_2(s) - D, \quad U(s) = \log b_1(s) + D.$$

The function N is strictly increasing. Put

$$A(s) = M(s) + \sup_{t \geq s_0} \{L(t) - M(t) - (N(t) - N(s))_+\}.$$

The supremum is finite and $L \leq A \leq U$: the required pairwise inequalities are exactly (4.11). As a function of $N(s)$, the supremum is nondecreasing and 1-Lipschitz. Consequently $m \leq A' \leq \tau' - m$ almost everywhere. Put $B = \tau - A$, $a = e^A$, $b = e^B$. Then $a \leq e^D b_1$, $b \leq e^D b_2$, $ab = e^s T$, and both logarithmic slopes are at least m.

It remains to smooth without losing these useful properties. Extend A, B to $s < s_0$ with slopes $1/2$ and extend T constantly. Convolve A and B with the same even probability kernel

$$k(s) = Z^{-1} e^{-\sqrt{1+s^2}}, \qquad |k^{(q)}| \leq C_q k.$$

Write $\widetilde{A} = k * A$, $\widetilde{B} = k * B$ and $\widetilde{T} = \exp[k * \log T]$. Since k has mean zero, $\widetilde{A} + \widetilde{B} = s + \log \widetilde{T}$. The original logarithmic slopes are nonnegative and bounded. Hence convolution changes A, B by bounded constants, preserves their width upper bounds up to fixed factors, and gives

$$|(\widetilde{A}')^{(q)}| \leq C_q \widetilde{A}', \qquad |(\widetilde{B}')^{(q)}| \leq C_q \widetilde{B}'.$$

The Lipschitz property of T implies $|\widetilde{T} - T| \leq C\eta$ and $|\widetilde{T}^{(q)}| \leq C_q \eta$. For completeness, these estimates follow by splitting the exponentially weighted convolution at $|t| = T(s)/(2\eta)$; on the inner part use $T(s-t) = T(s) + O(\eta|t|)$, and on the outer part use $T \geq K$ and the exponential kernel tail. Derivatives of $\log \widetilde{T}$ are obtained by convolving $(\log T)'$ with derivatives of k. Also

$$\widetilde{A}'(s) \geq \varepsilon e^{-c_1 T(s)} \int k(t) e^{-c_1 \eta |t|} \, dt \geq \varepsilon_* e^{-c_* \widetilde{T}(s)},$$

and similarly for $\widetilde{B}'$. The logarithmic upper bound on T is preserved by convolution. Dropping tildes proves the lemma.

For the quantitative spatial construction one may, uniformly in a mesh parameter m, include the additional cap

$$T_{\mathrm{end},m}(s) = K + \eta \sqrt{1 + (s - \log m)^2}$$

when taking the minimum defining T. Its exponential clock has a uniformly bounded integral, so every Hall estimate and every constant above is uniform in m. The smoothing changes this upper bound by at most a fixed additive constant. In particular, for $1 \leq u \leq C_1 m$,

$$e^{C T_m(\log u)} \leq C' \, (m/u)^{C'\eta}, \tag{4.12}$$

where the constants are independent of m. At the left endpoint, $T_m(s_0) \leq T_0(s_0)$; the two width bounds and their product therefore also give fixed positive lower bounds on the initial squared semiaxes. □

4.3 A rounded-rectangle Bellman supersolution

We use the functions supplied by Lemma 4.1, with $s = \log u$. Thus a, b are squared semiaxes and

$$ab = uT, \quad \alpha = D\log a, \quad \beta = D\log b, \quad \gamma = D\log T, \quad \alpha+\beta = 1+\gamma, \qquad D = u\frac{d}{du}.$$

They satisfy, uniformly in the mesh parameter when it is present,

$$\begin{aligned} T \geq K, \qquad |D^q T| \leq C_q\eta, \qquad \alpha, \beta \geq \delta := \varepsilon_* e^{-c_* T}, \\ |D^q\alpha| \leq C_q\alpha, \quad |D^q\beta| \leq C_q\beta, \qquad T \leq C\log(e+\log u). \end{aligned}$$

The small parameter η is chosen after the fixed constants below; the lower bound K can be increased as described in the allocation construction. Choose $\Lambda > c_* + 1$ and set

$$P = \Lambda T, \quad \lambda = P/a, \quad \mu = P/b, \quad X = \lambda x^2, \quad Y = \mu y^2, \quad E = e^X, \quad F = e^Y.$$

For each u, consider the convex oval

$$K_u = \{(x,y) : e^{\lambda x^2} + e^{\mu y^2} \leq e^P + 1\}. \tag{4.13}$$

Its intercepts are exactly $\sqrt{a}$ and $\sqrt{b}$. Write

$$G(s,x,y) = E + F - e^P - 1, \quad V = EX + FY, \quad D_0 = e^P P - EX - FY.$$

All identities below are evaluated on $G = 0$ unless indicated otherwise.

Lemma 4.2 (Rounded-rectangle phase)**.** *For sufficiently small η and sufficiently large K, the sets K_u are strictly nested. Their inverse level function $H(x,y) = u$ is smooth outside any fixed initial oval. With $g = \nabla H$, there is an explicit positive matrix $\mathcal{B}$ such that*

$$D^2H + gg^{\mathsf{T}} \preceq \mathcal{B}, \qquad \sqrt{\det\mathcal{B}} \leq C\,(xH_x + yH_y) \tag{4.14}$$

at all sufficiently high levels. The constants are uniform in the mesh parameter. After a fixed smooth flattening at low levels, isotropic spatial dilation gives a smooth phase $\mathcal{F}(t,z)$ and a positive matrix proxy $\mathcal{A}(t,z)$ with

$$D^2\mathcal{F} + \nabla\mathcal{F}\nabla\mathcal{F}^{\mathsf{T}} \preceq \mathcal{A}, \qquad -\mathcal{F}_t \geq \sqrt{\det\mathcal{A}}. \tag{4.15}$$

Its terminal cost controls both desired complementary Young modulars:

$$\psi_1(|x|/C) + \psi_2(|y|/C) \leq Ce^{\mathcal{F}(T,x,y)}. \tag{4.16}$$

The phase diverges at the boundary of its eventual domain.

Proof. First note the entropy identity

$$Q := -G_s = \alpha EX + \beta FY + \gamma D_0. \tag{4.17}$$

Indeed $D\log\lambda = \beta - 1 = \gamma - \alpha$ and $D\log\mu = \alpha - 1 = \gamma - \beta$. Since $E + F = e^P + 1$ and $E, F \geq 1$,

$$0 \leq D_0 \leq P\min(EX, FY). \tag{4.18}$$

For example, fixing P and taking $E \leq F$, the deficit vanishes at $E = 1$ and its derivative with respect to E is $\log(F/E) \leq P$. Thus $D_0 \leq P(E-1) \leq PEX$; exchange the coordinates for the other bound. Because $|\gamma|P = \Lambda|DT|$ can be made arbitrarily small,

$$Q \asymp \alpha EX + \beta FY, \qquad Q \geq c\{\min(EX, FY) + \delta V\} > 0. \tag{4.19}$$

This proves strict nesting and the existence of the inverse level function.

Here is a matrix upper bound that does not assume convexity of H. Put $p = \nabla_z G$ and

$$\mathcal{D} = \operatorname{diag}(G_{xx}, G_{yy}) = \operatorname{diag}\{2\lambda E(1+2X), 2\mu F(1+2Y)\}.$$

Implicit differentiation gives

$$g = \frac{u}{Q}p, \qquad D^2H + gg^{\mathsf{T}} = \frac{u}{Q}\mathcal{D} + \frac{u}{Q^2}(p \otimes G_{sz} + G_{sz} \otimes p) + \left\{ \frac{u(G_{ss}+Q)}{Q^3} + \frac{u^2}{Q^2} \right\} pp^{\mathsf{T}}. \tag{4.20}$$

The derivatives that enter this expression obey

$$G_{sx} = -(\alpha-\gamma)(1+X)p_1, \quad G_{sy} = -(\beta-\gamma)(1+Y)p_2, \tag{4.21}$$

$$G_{sz}^{\mathsf{T}}\mathcal{D}^{-1}G_{sz} \le CPV, \qquad |G_{ss}+Q| \le CPV.$$

For the last bound, differentiate the defining exponentials twice; $|DX| \le CX$, $|DY| \le CY$, $|DP| + |D^2P| \le C$, and $V \asymp Pe^P$. These estimates also follow directly from (4.21) and the stated logarithmic derivative bounds. Cauchy–Schwarz in the metric $\mathcal{D}$ now proves the Loewner bound in (4.14) with

$$\mathcal{B} = C\frac{u}{Q}\mathcal{D} + \left(C\frac{uPV}{Q^3} + \frac{u^2}{Q^2} \right) pp^{\mathsf{T}}. \tag{4.22}$$

Equivalently, it has the useful diagonal-plus-gradient form

$$\mathcal{B} = \operatorname{diag}(A_1, A_2) + (1+r)gg^{\mathsf{T}}, \quad A_i = C\frac{u}{Q}G_{ii}, \quad r = C\frac{PV}{uQ}.$$

The same estimates with absolute values of the cross and rank-one coefficients also give $-A - rgg^{\mathsf{T}} \preceq D^2H \preceq A + rgg^{\mathsf{T}}$. Thus the later spatial argument has a bound for both signs of the Hessian, not just for the exponential Hessian.

We give the determinant bound explicitly. The elementary identities are

$$R_0 := p^{\mathsf{T}}\mathcal{D}^{-1}p = \frac{2XE}{1+2X} + \frac{2YF}{1+2Y} \asymp V/P, \qquad xH_x + yH_y = \frac{2uV}{Q}.$$

The rank-one determinant formula therefore gives

$$\frac{\det \mathcal{B}}{(xH_x + yH_y)^2} \le C\frac{G_{xx}G_{yy}}{V^2}\left[1 + \left(\frac{V}{Q}\right)^2 + \frac{uV}{PQ} \right]. \tag{4.23}$$

The crucial mixed-exponential estimate is

$$EF(1+X)(1+Y) \le CVQ. \tag{4.24}$$

To check it, take $X \le Y$. Then $Y \asymp P$, $F \asymp e^P$, and $V \asymp PF$. If $X \ge 1$, the left side is at most $C(EX)(PF)$ and the $\min(EX, FY)$ term in (4.19) suffices. If $X \le 1$, the left side is at most $CPF \asymp V$, whereas $Q \ge c\delta V \ge c\varepsilon_* Pe^{(\Lambda - c_*)T}$. The last expression is bounded below by a sufficiently large positive constant after increasing K. This proves (4.24).

Finally

$$G_{xx}G_{yy} = \frac{4P^2}{ab}EF(1+2X)(1+2Y), \qquad ab = uT, \qquad V/Q \le C\delta^{-1}.$$

Substitution into (4.23) gives

$$\frac{\det \mathcal{B}}{(xH_x + yH_y)^2} \le C\left[1 + \frac{Te^{c_* T}}{u}\right] \le C. \tag{4.25}$$

The last inequality uses $T = O(\log(e + \log u))$. This proves the asserted high-level matrix bound.

For clarity, low-level extension costs only fixed constants. Fix a large u_0 and set $H = u_0$ inside K_{u_0}. Choose a smooth nondecreasing convex χ that vanishes on $(-\infty, u_0]$ and has $\chi' = 1$ above $u_0 + 1$. The function $\chi(H)$ is smooth, including across the initial oval. A positive proxy for its exponential Hessian is

$$\mathcal{B}_\chi = \chi' \mathcal{B} + (\chi'' + \chi'^2) g g^{\mathsf{T}}.$$

On the fixed transition annulus, all entries and their derivatives are bounded uniformly in m: the allocated widths there have fixed upper and positive lower bounds, and so do the slope and nesting constants. For the spatial selector use a fixed positive diagonal majorant cI throughout $H \le u_0 + 2$, with c large enough to dominate the absolute Hessian of $\chi(H)$ there. Its gradient rank term has coefficient one. Blend smoothly to the displayed high-level proxy only on $u_0 + 2 \le H \le u_0 + 3$, where $\chi' = 1$; the constant c also dominates on this fixed annulus. Convex combinations preserve the matrix upper bound. This gives a representation $A + (1 + r)\nabla\chi(H) \otimes \nabla\chi(H)$ with a uniformly positive compact diagonal A and a bounded r. In particular one does not divide χ'' by χ'^2 near the flat region. Above the annulus its determinant has the same bound as (4.25), up to a fixed factor.

Let $s(t) > 0$ satisfy $s(t)s'(t) = C_1$, with $s(0) = 1$, and put

$$\mathcal{F}(t, z) = \chi(H(z/s(t))) + C_0(T - t).$$

Choose C_1 sufficiently large to dominate the high-level constant in (4.14); choose C_0 to dominate the bounded transition annulus. Spatial dilation multiplies the positive matrix proxy by $s(t)^{-2}$, while

$$-\mathcal{F}_t = \frac{s'}{s}\chi'(H)\, z/s \cdot \nabla H + C_0.$$

This yields (4.15). A small isotropic matrix regularization, with a fixed additional time margin, makes the selector strictly positive even in the flat region.

Every level obeys $|x| \le \sqrt{a(u)}$ and $|y| \le \sqrt{b(u)}$. The allocation bounds consequently give $\psi_i(|z_i|/C) \le e^u$ on that level. Fixed spatial dilation and the identity $\chi(u) = u - O(1)$ prove (4.16). The finite initial and transition levels are absorbed in the same fixed constant. As usual, convexity of ψ_i turns a fixed bound on its modular into a Luxemburg bound after one further fixed dilation. Strict nesting of compact levels also shows that the phase diverges at the boundary of their union. □

The formulas also display the quantitative bounds needed for spatial realization. Differentiating a fixed number of times in logarithmic level and in the normalized coordinates $\sqrt{a}\,\partial_x$, $\sqrt{b}\,\partial_y$ introduces only fixed powers of u and e^{CT}. The logarithmic growth cap gives polylogarithmic losses in the level, and the mesh-end cap gives the sharper estimate

$$e^{CT_m(\log u)} \le C(m/u)^{C\eta} \qquad (u \le C_1 m).$$

Thus choosing η sufficiently small preserves the small conditional step and oscillation errors in the Gaussian realization.

5 Spatial realization of the phase

5.1 Spatial realization of the two-cost phase

We give the spatial argument, including the rank-one Hessian term and the exact repairs for both Young norms. Write $u = H(z/s(t))$ for the underlying implicit level and $H_m(t,z) = \chi(u)$ for its flattened phase. We keep the bounded time-only correction separate, in the modular below. Above the cutoff annulus H_m differs from u by one fixed additive constant, so their spatial and time derivatives agree. All high-level formulas below use this underlying u. Put $v = \nabla H_m$ and $A = \operatorname{diag}(d_1, d_2)$. The time-dilation margin Λ_{time} below is independent of the already fixed shape coefficient in $P = \Lambda T$. Set

$$B = A + vv^{\mathsf{T}}, \quad K = \sqrt{\det B}, \quad C = KB^{-1}, \quad b_i^2 = K/d_i.$$

These are deliberately the slow curvature widths, rather than the diagonal entries of the selector. On the whole compact flattened region choose a fixed sufficiently large diagonal matrix $A = cI$ dominating both signs of its Hessian, and put $r = 0$. Blend to the high-level formula only in a fixed annulus above the point where the cutoff derivative has become one. All coefficients on this annulus are uniformly bounded, so this preserves a bounded r and avoids division by a vanishing cutoff derivative. The preceding phase calculations supply the following estimates above the fixed cutoff annulus, with $u \le C_1 m$:

$$-A - rvv^{\mathsf{T}} \preceq D^2 H_m \preceq A + rvv^{\mathsf{T}}, \qquad 0 \le r \le C, \quad r \le u^{-1} e^{CT_m(\log u)}, \tag{5.1}$$

$$K \le u e^{CT_m(\log u)}, \qquad v_i^2/d_i \le u e^{CT_m(\log u)}, \qquad -\partial_t H_m \ge c\Lambda_{\text{time}} K, \tag{5.2}$$

$$|b_i \partial_i \log f| \le e^{CT_m(\log u)}, \qquad |\partial_t \log f| \le \Lambda_{\text{time}} e^{CT_m(\log u)} \quad (f = 1+u, d_1, d_2, K, b_1, b_2), \tag{5.3}$$

$$|\partial_t v_i| \le \Lambda_{\text{time}} e^{CT_m(\log u)} |v_i|. \tag{5.4}$$

The finitely many differentiated comparison estimates used below have the same form after increasing C. All widths, coefficients, and reciprocals have polynomial bounds in m; the flat compact region has uniform constant versions. The negative time derivative estimate is only asserted above the cutoff annulus; the separate scalar clock will cover its complement. For comparisons across the compact region use the globally smooth quantity $1 + H_m$, which is comparable to u at high levels, in place of $1 + u$. The mesh cap ensures, for every fixed power required in these estimates,

$$e^{CT_m(\log u)} \le C'(m/u)^{C'\eta_{\text{path}}} \qquad (1 \le u \le C_1 m). \tag{5.5}$$

The path parameter is chosen small after the finite powers are fixed.

Here is a precise order for these parameter choices. First fix the prefix-data constants, the width regularization exponent, the positive smoothing kernel and its finitely many derivative orders, the clock exponent c_*, and the shape parameter in $P = \Lambda T$. The exponential powers denoted C are fixed at this stage: all estimates are fixed-degree polynomials in P, the inverse slope lower bound $\delta_{\text{slope}}^{-1} = \varepsilon_*^{-1} e^{c_* T}$, and the relative slope derivatives. Dependence of the Hall padding or initial axes on later parameters enters only multiplicative constants. Choose η_{path} using these fixed exponential powers and η_{clip}. Then choose K sufficiently large, of order at least $\log(1/\eta_{\text{path}})$, to suppress the clock convolution losses, and choose the fixed cutoff u_0 large enough to absorb all remaining compact prefactors. The compact diagonal proxy and scalar clock may depend on these choices. The metric-memory coefficient, time-drift margin and finally ε are chosen afterwards. This separates exponent choices from compact prefactors and removes any apparent circularity in the mesh-cap argument.

The slow curvature widths satisfy the exact identities

$$K^2 = d_1 d_2 + d_1 v_2^2 + d_2 v_1^2, \qquad C_{ii} = (d_j + v_j^2)/K \le b_i^2, \qquad d_i b_i^2 = K \quad (j \ne i). \tag{5.6}$$

Write $U_* = v_1^2/d_1 + v_2^2/d_2$. Then $b_1^2 b_2^2 = 1 + U_*$. For determinant-one G with $G_{ii} \le M_0 b_i^2$, put $\mathcal{L}(G) = \sqrt{\operatorname{tr}(GC^{-1})}$. Then

$$\mathcal{L}^2 \lesssim_{M_0} 1 + U_* \lesssim_{M_0} (1+u) e^{CT_m(\log(1+u))}, \qquad v^{\mathsf{T}} G v \le K \mathcal{L}^2. \tag{5.7}$$

Indeed $\operatorname{tr}(GA) \le 2M_0 K$, while $v^{\mathsf{T}} G v \le 2M_0 K U_*$.

Here is an explicit verification of the scalar derivative bounds, which is useful near the coordinate axes. In the notation of the rounded phase let $\mathcal{D} = \operatorname{diag}(G_{xx}, G_{yy})$, $p = \nabla_z G$, and $R_0 = p^{\mathsf{T}} \mathcal{D}^{-1} p \asymp V/P$. Since $d_i = c(u/Q)\mathcal{D}_{ii}$ for a fixed positive c,

$$K^2 = (u/Q)^2 \det \mathcal{D}\, [c^2 + c(u/Q) R_0].$$

Here $V \asymp P e^P$ and $R_0 \asymp e^P$ on every level. Thus neither logarithmic differentiation of R_0 nor that of the bracket introduces a small coordinate-gradient denominator. The formulas also give

$$e^{-CT_m} \le b_1^2/a,\ b_2^2/b \le e^{CT_m}.$$

For this estimate use $\mathcal{D}_{11} = 2(P/a)E(1+2X)$, $\mathcal{D}_{22} = 2(P/b)F(1+2Y)$, $ab = uT$, $R_0/Q \ge c/P$, and $R_0/Q \le e^{CT_m}/P$. In the semiaxis directions $\sqrt{a}\,\partial_x$ and $\sqrt{b}\,\partial_y$, implicit differentiation gives $|\partial \log u| \le e^{CT_m}$, since $Q \ge c\delta V$. The identities $Q_s = -G_{ss}$ and $Q_{z_i} = -G_{sz_i}$, together with $|G_{ss}| \lesssim PV$ and the displayed mixed derivatives of G, bound derivatives of $\log Q$ by e^{CT_m}. The explicit positive formulas for $\mathcal{D}$, R_0, and K^2 then prove (5.3). Convert to the slow widths by their comparison with the semiaxes. Repeated differentiation introduces only fixed powers of $P, \delta_{\text{slope}}^{-1}$ and their controlled logarithmic derivatives, so the same assertion holds through every fixed required order. For the time derivatives use isotropic dilation and $u_t/u = O(\Lambda_{\text{time}} V/Q)$; each nonzero v_i is a coordinate z_i times a positive coefficient with logarithmic derivative bounded by $\Lambda_{\text{time}} e^{CT_m}$. A zero coordinate remains zero under dilation. This proves (5.4).

Use the rectangular factor S_C with rows $\sqrt{d_1/K}\, e_1^{\mathsf{T}}$, $\sqrt{d_2/K}\, e_2^{\mathsf{T}}$, and $v^{\mathsf{T}}/\sqrt{K}$. Thus $S_C^{\mathsf{T}} S_C = C^{-1}$ and $\mathcal{L} = \|S_C G^{1/2}\|_{\text{HS}}$. The first two rows after multiplication by $G^{1/2}$ are bounded, by (5.6). Their derivatives use (5.3). For the last row use (5.1): with $D = A + rvv^{\mathsf{T}}$, $|D^2 H_m(w,z)| \le \sqrt{D(w,w)D(z,z)}$. Consequently

$$\begin{aligned} \|b_i \partial_i (S_C G^{1/2})\|_{\text{HS}} &\lesssim_{M_0} \sqrt{1+u}\, e^{CT_m(\log(1+u))}, \\ \|\partial_t (S_C G^{1/2})\|_{\text{HS}} &\lesssim_{M_0} \Lambda_{\text{time}} \sqrt{1+u}\, e^{CT_m(\log(1+u))}. \end{aligned} \tag{5.8}$$

For example the square of the differentiated-gradient contribution is at most $K^{-1} D(b_i e_i, b_i e_i) \operatorname{tr}(GD)$, and both factors are at most K times a fixed power of e^{CT_m}. The time estimate uses (5.4). This factor calculation avoids small eigenvalue denominators.

Proposition 5.1 (Two-cost spatial realization)**.** *The phase with these quantitative estimates yields*

$$\mathbb{E} \inf_{\operatorname{div} J = P_h W^0,\ J\cdot n = 0} \max_{i=1,2} \|J_i\|_{L^{\psi_i}} \le C\sqrt{\log(1/h)}.$$

Proof. Put $m = \log(1/h)$ and fix $0 < \eta_{\text{clip}} < 1/10$. Choose η_{path} so small that all exponents $C\eta_{\text{path}}$ below are less than η_{clip} and $1/4$. Set

$$\delta = \varepsilon^2/m, \qquad E_0 = m^{1/2 - \eta_{\text{clip}}}, \qquad s = h m^{C_s},$$

with C_s large fixed. Write $t_k = k\delta$ and start from $U_0 = 0, G_0 = I$. At an active parent Q choose an old-measurable interior representative x_Q and set $C_Q = C(t_k, U_k(x_Q))$. If

$$G^{-1/2} C_Q G^{-1/2} = O \operatorname{diag}(\lambda, \lambda^{-1}) O^{\mathsf{T}}, \qquad \lambda^{\#} = \min\{K_0, \max\{K_0^{-1}, \lambda\}\},$$

put

$$G_+ = G^{1/2} O \operatorname{diag}(\lambda^{\#}, (\lambda^{\#})^{-1}) O^{\mathsf{T}} G^{1/2}.$$

Thus the two reciprocal relative eigenvalues are clipped reciprocally. The scalar inequalities $K_0^{-1} \le \lambda^{\#} \le K_0$ and $\lambda^{\#} \le K_0^{-1} + \lambda$ give

$$\det G_+ = 1, \quad K_0^{-1} G \preceq G_+ \preceq K_0 G, \qquad G_+ \preceq K_0^{-1} G + C_Q.$$

Take the ordinary Whitney split in this chart down to minimum leg s. Apply its local ray inverse, multiply the current by $\sqrt{\delta}$, and use even coarse clipping at E_0. Write ΔU_j for the retained increment born at generation j, in metric $G_j = A_j A_j^{\mathsf{T}}$, and ℓ_j for its child leg. On that birth child its envelope E_j satisfies

$$\|A_j^{-1} \Delta U_j\|_\infty \le C\sqrt{\delta} E_j, \qquad |A_j^{-1}(\Delta U_j(x) - \Delta U_j(x'))| \le C\sqrt{\delta} E_j \left(\frac{|A_j^{-1}(x - x')|}{\ell_j} \right)^{\zeta},$$

for one fixed $0 < \zeta < 1$. Each retained E_j is at most CE_0 and has a uniform conditional Gaussian tail. Freeze geometric terminal cells. The depth is at most $C_* m$; take horizon $T = C_* m \delta$ and eventually choose ε so all phase time restrictions hold. Stop the whole modular at its first crossing of a large fixed constant.

Here is the full geometric bootstrap. Inductively maintain the pointwise diagonal cap $G_{k,ii} \le M_0 b_i(t_k, U_k)^2$, validity of the phase, and per-generation width ratios in $[1 - c_0, 1 + c_0]$, for a fixed sufficiently small $c_0 > 0$. These hold initially. If a current parent is a descendant of a birth child at generation $j \le k$, its diameter divided by that birth leg, in the birth metric, is at most $C\vartheta^{k-j}$. Indeed consecutive charts have relative distortion at most $\sqrt{K_0}$; choose the intrinsic Whitney contraction so small that its product with this fixed distortion is $\vartheta < 1$.

For x, x_0 in the same current parent, use the birth cap at x_0 and the width-ratio induction along that fixed spatial point:

$$\begin{aligned} |\Delta U_{j,i}(x) - \Delta U_{j,i}(x_0)| &\le C\sqrt{\delta}\, \sqrt{G_{j,ii}}\, E_j(x_0) \vartheta^{\zeta(k-j)} \\ &\le C\sqrt{\delta}\, b_i(t_k, U_k(x_0)) \big[(1 - c_0)^{-1} \vartheta^{\zeta}\big]^{k-j} E_j(x_0). \end{aligned}$$

Set $r_{\mathrm{osc}} = (1 - c_0)^{-1} \vartheta^{\zeta} < 1$ and fix $r_{\mathrm{osc}}^2 < \rho < 1$. Define, pointwise along the actual birth history,

$$\mathcal{T}_k(x) = \sum_{j \le k} \rho^{k-j} E_j(x)^2, \qquad \mathcal{T}_{k+1} = \rho \mathcal{T}_k + E_{k+1}^2.$$

Thus $\mathcal{T}_k \le C E_0^2$. Summing the displayed birth estimate and using $\sum_{\ell \ge 0} (r_{\mathrm{osc}}^2/\rho)^{\ell} < \infty$ gives

$$|U_{k,i}(x) - U_{k,i}(x_0)| \le C\sqrt{\delta}\, b_i(t_k, U_k(x_0)) \sqrt{\mathcal{T}_k(x_0)}. \tag{$*$}$$

This deterministic estimate is anchored at x_0; it does not presuppose any bound for the phase at the other point.

Before a modular stop each active cell has a witness x_0 with $1 + H_m(t_k, U_k(x_0)) \le C_1 m$, because its area is at least cs^2 and its integral of e^{H_m} is bounded by the modular cap. The witness need

not be chosen predictably: it is used only in the deterministic bootstrap. Integrate the coefficient derivative bounds along the straight state segment from $U_k(x_0)$ to $U_k(x)$. Up to the first possible failure of factor-two bounds for $1 + H_m, d_i, K, b_i$, estimate (*) gives relative logarithmic change at most

$$C\sqrt{\delta}E_0 e^{CT_m} \le C\varepsilon m^{-\eta_{\mathrm{clip}}+C\eta_{\mathrm{path}}} = o(1).$$

It prevents that first failure and hence continues to every x. A state segment cannot escape the phase domain while this comparison holds: properness of the nested levels would force H_m to tend to infinity. Thus all old states have $H_m \le 2C_1 m$ and their widths and coefficients are relatively $1 + o(1)$ throughout the parent.

At the representative the new metric satisfies $G_{+,ii} \le (K_0^{-1}M_0 + 1)b_i^2$. Choose M_0 so large that $K_0^{-1}M_0 + 1 \le (1 - 2c_1)M_0$ for some $c_1 > 0$. The preceding representative comparison gives the same bound with margin c_1 at every old point. The fresh displacement divided by its old widths is at most $C\sqrt{M_0\delta}E_0$. The same first-failure argument along this fresh state segment and the short time interval δ gives relative coefficient change $o(1)$ and an upper level bound $C_1' m$. The new diagonal cap follows from its strict old margin. In particular the new width ratios lie in $[1 - c_0, 1 + c_0]$ for large m, closing the bootstrap used in (*). The coefficient bounds now give polynomial chart distortion.

For completeness the lag contraction is also explicit. If e^d is the larger relative eigenvalue of G against a fixed C_Q, then $\mathcal{L}(G)^2 = 2\cosh d$. Clipping replaces d by $(d - \log K_0)_+$, so $\mathcal{L}(G_+)^2 \le K_0^{-1}\mathcal{L}(G)^2 + 2$. Representative and fresh-state comparisons cost, by (5.8), at most

$$C\sqrt{\delta(1+u)}e^{CT_m}(\sqrt{\mathcal{T}} + E) + C\Lambda_{\mathrm{time}}\delta\sqrt{1+u}e^{CT_m} \le C\varepsilon(\sqrt{\mathcal{T}} + E) + C\Lambda_{\mathrm{time}}\varepsilon^2.$$

Consequently, with $q_1 < 1$ fixed,

$$\mathcal{L}' \le q_1\mathcal{L} + C + C\varepsilon(\sqrt{\mathcal{T}} + E) + C\Lambda_{\mathrm{time}}\varepsilon^2.$$

Choose ε after Λ_{time}. Squaring with a fixed slack in $q_1 < 1$ gives $\mathcal{L}'^2 \le q_2\mathcal{L}^2 + C + C\varepsilon^2(\mathcal{T} + E^2)$ with $q_2 < 1$. Adding $\mathcal{T}' = \rho\mathcal{T} + E^2$ and taking ε small gives, for $S = \mathcal{L}^2 + \mathcal{T}$ and $q < 1$,

$$S' \le qS + C(1 + E^2), \qquad K'S' - KS \le -cKS + CK(1 + E^2). \tag{5.9}$$

The relative change of K is uniformly $o(1)$.

Use $\Xi = H_m(t, U) + C_0(T - t) + \kappa\delta KS$. The predictable linear tilt has Gaussian variance at most $C\delta K\mathcal{L}_+^2$; even scalar clipping decreases its exponential moment. The rank-one Hessian term requires a scalar comparison; the gradient itself is not assumed comparable along a segment. Write ξ for the retained increment and $f(s) = H_m(t, U + s\xi)$. The proved comparisons of A, K, b_i give

$$f''(s) \le a + \widehat{r}f'(s)^2, \qquad a = C\delta KE^2, \qquad \widehat{r} = C(1+u)^{-1}e^{CT_m(\log(1+u))}.$$

Here $\xi^{\mathsf{T}}A\xi \le a$ follows from the diagonal cap and $d_i b_i^2 = K$. Moreover $|f'(0)| \le C\sqrt{\delta K}\,\mathcal{L}_+ E$. The mesh cap and $E \le CE_0$ imply $\widehat{r}|f'(0)| + \widehat{r}a = o(1)$: the square of the first term and the second term are bounded by a fixed power of e^{CT_m} times $C\delta E_0^2 = o(1)$. Scalar Riccati comparison therefore gives

$$f(1) - f(0) \le f'(0) + Ca + C\widehat{r}f'(0)^2.$$

Indeed compare $z = \widehat{r}f'$ to $z' = \widehat{r}a + z^2$; small initial value and forcing bound its increment by $C(\widehat{r}a + z(0)^2)$. The case $\widehat{r} = 0$ is immediate. The spatial Taylor contribution is thus at most $C\delta K(1 + \widehat{r}\mathcal{L}_+^2)E^2$. Use $\widehat{r}$ in place of r in the following estimates; it has the same uniform and mesh-floor bounds.

The moment bounds remain uniform. Besides $\delta K \lesssim \varepsilon^2$, the mesh cap yields

$$\delta r K \mathcal{L}^2 \lesssim \delta u e^{CT_m} \lesssim \varepsilon^2, \qquad \delta r K \mathcal{T} \lesssim \varepsilon^2 m^{-2\eta_{\mathrm{clip}}+C\eta_{\mathrm{path}}} \lesssim \varepsilon^2.$$

Thus the conditional envelope moment and Cauchy–Schwarz bound the positive quadratic contribution by $C\delta K + C\delta r K S$ in the logarithmic moment. The phase gives negative time drift $-c\Lambda_{\mathrm{time}}\delta K$; the scalar clock covers its compact flat and transition regions. Together with (5.9), the logarithmic moment of $\Xi' - \Xi$ is at most

$$-c\Lambda_{\mathrm{time}}\delta K - c\kappa\delta K S + C\delta K\mathcal{L}^2 + C\delta r K S + C_\kappa \delta K.$$

Since r is globally bounded, choose κ to pay the two terms proportional to S, then $\Lambda_{\mathrm{time}}, C_0$ to pay the constant, and ε last. Spatial integration with prescribed freezing gives a nonnegative supermartingale with bounded constant initial value. Its maximal inequality gives a fixed positive probability of completing without a modular stop. The final modular and the phase's level-set inclusion imply $\|U_i\|_{L^{\psi_i}} \le C$ for both i.

We restore the exact divergence using both suprema in the sparse repair proof. With $\sigma = \sqrt{\delta}$, $T_0 = \sigma\sqrt{m} = \varepsilon$, and $D_0 = m/E_0^2 = m^{2\eta_{\mathrm{clip}}}$, its support has area at most $e^{-cE_0^2}$ and

$$\|J_1\|_\infty \le CT_0D_0B, \qquad \|J_2\|_\infty \le CT_0D_0/B.$$

Allow $B < 1$ by exchanging the axes. Put $R_i = \rho_i(e^{c'E_0^2})$ for small fixed $c' > 0$ and choose

$$B = \sqrt{\widehat{R}_1/\widehat{R}_2}, \qquad \widehat{R}_i = \min\{R_i, m\}.$$

The prefix condition implies $R_1R_2 \gtrsim E_0$. To recall the short proof, if $p_j = \rho_1(e^{2^j})^2\rho_2(e^{2^j})^2\,2^{-j}$ tended to zero, then for all $j - r \le k \le j$ monotonicity would give $\rho_1(e^{2^k})^2\rho_2(e^{2^j})^2\,2^{-k} \le 2^r p_j$. Taking r proportional to $\log(1/p_j)$ produces arbitrarily many terms bounded below in the prefix sum, a contradiction. There are enough preceding indices because both inverse widths have fixed positive lower bounds. Thus $\inf_j p_j > 0$; interpolation between dyadic levels gives $\rho_1(e^v)\rho_2(e^v) \gtrsim \sqrt{v}$. Each R_i is bounded below by a fixed positive constant, so truncating at m still leaves $\widehat{R}_1\widehat{R}_2 \gtrsim E_0$, since $E_0 \le \sqrt{m}$. The ratio B is polynomially bounded in both directions. A field of supremum L on area p has norm at most $L/\rho_i(1/p)$. Both normalized repair norms are therefore at most

$$CT_0D_0/\sqrt{E_0} = O(m^{(5/2)\eta_{\mathrm{clip}}-1/4}) = o(1).$$

This argument also covers costs stronger than Gaussian; it does not use a false ψ_2 embedding.

Here are the exact terminal corrections, with normalization made explicit. The preceding retained currents and clipping repairs have divergence $\sqrt{\delta}f_{\mathrm{leaf}}$. Thus f_{leaf} denotes the *raw* revealed Gaussian density, before multiplication by $\sqrt{\delta}$. At each generation there are at most Cs^{-2} attempted cells, because every child has area at least cs^2. The number over all generations is at most $Cms^{-2} = \exp(O(m))$. Every fresh cell density has conditional standard deviation at most C/s. Gaussian tails with threshold $C_A\sqrt{m}/s$, followed by the union bound, give simultaneous density bounds with failure at most e^{-Am}, where A is arbitrarily large by increasing C_A. Summing at most C_*m ancestral contrasts proves

$$\|f_{\mathrm{leaf}}\|_\infty \le Cm^{3/2}/s$$

with probability $1 - o(1)$. This estimate concerns the full revealed contrasts: the sparse repairs have already restored the divergences removed by clipping.

The raw correction $(P_h - I) f_{\mathrm{leaf}}$ has mean zero in every h-square. Row primitives after subtracting row means, followed by the vertical primitive of those means, give zero-normal currents in that square of both suprema at most $Ch\|f_{\mathrm{leaf}}\|_\infty = O(m^{3/2-C_s}) = o(1)$. The corresponding normalized correction is smaller by $\sqrt{\delta}$, although we only need the displayed raw bound.

By the geometric stopping rule and polynomial chart cap, all terminal leaves have physical diameter at most hm^{C_2} for a fixed C_2. Choose an h-aligned dyadic length D satisfying

$$4(\max_Q \operatorname{diam} Q + \sqrt{2}h) \le D \le Chm^{C_2};$$

for large m it is less than $1/4$. Partition the unit square into dyadic D-squares and assign each terminal leaf to the square containing one of its representatives. Its final-grid projection is supported in that square and its immediate neighbors. Enclose these supports in an h-aligned square of side $4D$; at the domain boundary slide that square inside the unit square. It can contain every assigned support, because their coordinate spans are at most $3D$. Each enlargement lies within $O(D)$ of its original grouping square, so the enlarged boxes have a universal bounded overlap.

Condition on the full revealed history. For a grouping box R, the unresolved source is the sum of the bridges on its assigned leaves. These are independent isonormal processes on the orthogonal spaces $L^2_0(Q)$ of the assigned leaves. If Π_R is their orthogonal-sum projection, its raw projected cell masses divided by h are

$$g_v = W\big(\Pi_R(\mathbf{1}_v/h)\big).$$

The tests $\mathbf{1}_v/h$ are orthonormal, and Π_R is a contraction. Thus the conditional covariance of (g_v) is at most the identity. Its sum is exactly zero: the grouping box contains every assigned leaf, and each leaf bridge annihilates its constant. The capacity lemma applies to the $4D/h$ square mesh. Its mean is $O(\log(D/h)) = O(\log m)$; concentration with deviation $C_A\sqrt{m}$ gives failure e^{-Am} for arbitrary fixed A, after choosing C_A. There are at most $h^{-2} = e^{2m}$ groups, so the union bound gives simultaneous raw capacity $O(\sqrt{m})$ with probability $1 - o(1)$. Face interpolation gives divergence g_v/h, exactly the required projected bridge density, and zero normal trace on each enlarged box. Bounded overlap then gives a sum with both raw suprema $O(\sqrt{m})$. These bounds control both Young norms after a fixed dilation. Their normalized costs would be $O(\sqrt{\delta m}) = O(\varepsilon)$.

Undo $\sqrt{\delta}$ and add every repair. The divergence is exactly $P_h W^0$ and the desired gauges hold with fixed positive probability. For clarity, the final expectation upgrade is also quantitative. For a mean-zero cell vector g, the grid cut formula and $|\partial S| \ge c\sqrt{\min(|S|, |S^c|)}$ give capacity at most $C\|g\|_2$ by Cauchy–Schwarz. Face interpolation produces a current of divergence $h^{-1}g$ and both suprema $C\|g\|_2$. A constant of size L has L^{ψ_i} norm $L/\rho_i(1)$, so the minimum-flow gauge $\mathcal{F}_h$ satisfies

$$|\mathcal{F}_h(g) - \mathcal{F}_h(g')| \le C_{\psi_1,\psi_2}\|g - g'\|_2.$$

The centering projection is a Euclidean contraction, and hence this Lipschitz constant is independent of the mesh for standard Gaussian histogram coefficients. The preceding construction gives $\mathbb{P}(\mathcal{F}_h \le C\sqrt{m}) \ge \varepsilon_0 > 0$ uniformly in m. The lower Gaussian concentration bound therefore implies

$$\mathbb{E}\mathcal{F}_h \le C\sqrt{m} + C_{\psi_1,\psi_2}\sqrt{2\log(1/\varepsilon_0)}.$$

This is the claimed estimate. The finitely many small values of m are absorbed by enlarging its constant. □

6 Necessity and exact examples

6.1 The entropy obstruction

We give the lower argument for the coordinate cost $\Psi(x,y)=\theta(|x|)+|y|$; exchanging the axes gives the convention used in the question. Here θ is finite and convex on $[0,\infty)$, $\theta(0)=0$, and $\theta(t)\ge t$. In particular its inverse is continuous. Put

$$a_j=\max\left\{1,\frac{e^{2^j}}{2^{j/2}\theta^{-1}(e^{2^j})}\right\}. \tag{L1}$$

Lemma 6.1 (Entropy criterion)**.** *For a sequence $a_j\ge 1$,*

$$\sup_{\substack{q_j\ge 0\\ \sum_j q_j\le 1}}\sum_j \frac{q_j\sqrt{\log(1/q_j)}}{a_j}<\infty \quad\Longleftrightarrow\quad \sum_j e^{-ca_j^2}<\infty \text{ for some } c>0.$$

The supremum may be restricted to finitely supported sequences.

Proof. For sufficiency choose A large and split each summand according as $\sqrt{\log(1/q_j)}\le Aa_j$ or not. The first sum is at most A. In the second, writing $x=\sqrt{\log(1/q_j)}>Aa_j$, $e^{-x^2}x/a_j\le C_A e^{-A^2a_j^2/2}$, which is summable for sufficiently large A.

Conversely suppose every such exponential series diverges. Fix $0<\gamma<1$, discard any finite initial segment, and, for each large A, select a finite set of remaining indices with

$$1\le Z_A:=\sum_j e^{-Aa_j^2}\le 2,\qquad q_j=\gamma e^{-Aa_j^2}/Z_A. \tag{L2}$$

This is possible by taking a partial sum up to its first crossing of one; each term is at most e^{-A}. Then $\sum q_j=\gamma$ and $\log(1/q_j)\ge Aa_j^2$, so the entropy sum is at least $\gamma\sqrt{A}$. □

Lemma 6.2 (A bounded-cost Brownian control)**.** *If the series in Lemma 6.1, with* (L1)*, diverges for every $c>0$, then for every $D>0$ there is a bounded predictable upper-triangular matrix process L_t, with nonnegative diagonal entries, such that its martingale $Z_t=\int_0^t L_s\,dB_s$ stays in a fixed compact set and*

$$\mathbb{E}\Psi(Z_1)\le 3,\qquad \mathbb{E}\int_0^1\sqrt{\det L_t}\,dt\ge D. \tag{L3}$$

The bounds on the process and its compact set may depend on D.

Proof. We first construct an uncapped process and subsequently stop it in the state variable. Set

$$S_j=2^j,\quad H_j=e^{8S_j},\quad \widetilde{a}_j=\max\{1,H_j/(\theta^{-1}(H_j)\sqrt{S_j})\}.$$

Since the quantity before taking the maximum is $\sqrt{8}$ times the corresponding quantity at index $j+3$ in (L1), the exponential series for $\widetilde{a}_j$ also diverges for every constant. Choose the finite probabilities q_j by (L2), using these $\widetilde{a}_j$ and only indices $j\ge j_0$, where j_0 will be fixed below. Take γ to be a sufficiently small absolute constant. Define

$$b_j=(\sqrt{S_j}\widetilde{a}_j)^{-1},\quad \ell_j=q_jb_j,\quad K_j=\log(1+q_j^{-2}),\quad w_j(v)=\frac{S_j}{K_j(v+\ell_j^2)}. \tag{L4}$$

In operational stage j, start a fresh real Brownian motion U at zero and stop at

$$T_j = \tau_{\ell_j} \wedge \tau_{\{\pm b_j\}} \wedge b_j^2, \qquad R_j = \int_0^{T_j} \sqrt{w_j(v)}\, dv,$$

where τ_ℓ is the first time Tanaka-normalized local time at zero reaches ℓ. Success means that τ_{ℓ_j} occurs first; at success $U_{T_j} = 0$. The following estimates have absolute constants, uniformly in all the parameters:

$$\mathbb{P}(\text{failure}) \le Cq_j, \qquad \mathbb{E}[R_j; \text{success}] \ge \frac{cq_j\sqrt{K_j}}{\widetilde{a}_j}, \qquad 0 \le R_j \le \frac{2}{\widetilde{a}_j\sqrt{K_j}}. \tag{L5}$$

For completeness, Tanaka's formula itself follows by applying Itô's formula to smooth convex approximations of $|x|$. It reads $|U_t| = M_t + L_t^0$, where $M_t = \int_0^t \operatorname{sgn}(U_s) dU_s$ is Brownian motion because its quadratic variation is t. The regulator L^0 increases only when $|U| = 0$, so the elementary reflection identity gives $L_t^0 = -\min_{s\le t} M_s$. Reflection at the first crossing of a positive level yields $\mathbb{P}(\tau_\ell > v) = 2\Phi(\ell/\sqrt{v}) - 1 \asymp \min(1, \ell/\sqrt{v})$. At the bounded stage stopping time, Tanaka's formula gives $\mathbb{E}|U_{T_j}| = \mathbb{E}L_{T_j}^0 \le \ell_j$, so exit at $\pm b_j$ has probability at most q_j. The reflection principle and Lévy's local-time identity give $\mathbb{P}(\tau_\ell > v) \asymp \min(1, \ell/\sqrt{v})$; the time-cap failure probability is therefore $O(q_j)$. Subtracting the total failure probability shows, for a sufficiently small absolute $c_0 > 0$,

$$\mathbb{P}(T_j > v, \text{success}) \ge c\ell_j/\sqrt{v}, \qquad \ell_j^2 \le v \le c_0 b_j^2.$$

Integration against $\sqrt{w_j(v)}\, dv$ proves the middle bound of (L5); direct integration proves the last one.

Conditioned on all stages succeeding, their resources are independent. Their sum has mean $\mu_A \ge c\gamma\sqrt{A}$. Moreover $\log(1/q_j) \ge A\widetilde{a}_j^2$, so its variance is at most $C\mu_A/\sqrt{A}$. The probability of all successes is at least $1 - C\gamma$. Consequently, for large A, with probability at least $1/2$, all stages succeed and their total resource exceeds $R := \mu_A/4$, where $R \to \infty$.

Run these stages using the first coordinate B^1 of a planar Brownian motion, with ordinary time given by

$$dt = \sqrt{w_j(v)}\, dv/R, \qquad \sigma_t^2 dt = dv, \quad \beta_t^2 dt = w_j(v) dv.$$

Stop at the first failed stage, the end of the finite stage list, or ordinary time one. Set $\sigma = \beta = 0$ afterward. During activity $\sigma\beta = R$. The stage clocks satisfy

$$\int_{\text{stage } j} \beta_t^2 dt \le S_j, \qquad s_t := \int_0^t \beta_r^2 dr \le 2S_j \quad \text{while visiting stage } j. \tag{L6}$$

The coefficients are bounded for this fixed finite list and depend only on B^1. The process $M = \int \sigma\, dB^1$ is confined to $[-b_j, b_j]$ during stage j, and restarts from zero after every success. Define

$$Y_t = \exp\left(\int_0^t \beta_r\, dB_r^2 - s_t/2\right), \qquad X_t = Y_t M_t.$$

Then the martingale $(X, Y - 1)$ has integrand

$$\mathcal{L}_t = \begin{pmatrix} Y_t\sigma_t & M_t Y_t \beta_t \\ 0 & Y_t\beta_t \end{pmatrix}, \qquad \sqrt{\det \mathcal{L}_t} = Y_t\sqrt{R} \quad \text{during activity.} \tag{L7}$$

Now stop at the first crossing $Y = H_j$ during stage j, calling this time τ. To estimate its cost in reward, use the *uncapped* probability measure $dQ = Y_1\, d\mathbb{P}$. Conditionally on the complete B^1 path,

the coefficients and clock are fixed, and under Q, $\log Y$ is a Brownian motion plus half its variance clock. Also the B^1 marginal is unchanged, since $\mathbb{E}(Y_1 \mid B^1) = 1$. By (L6) and reflection,

$$Q(\tau \le 1) \le \sum_{j \ge j_0} 2e^{-49 S_j/4}.$$

Choose j_0 so that this sum is at most $1/8$. The event of all successes and total resource at least R keeps the uncapped process active throughout $[0,1]$. Therefore

$$\mathbb{E}\int_0^{1\wedge\tau} \sqrt{\det \mathcal{L}_t}\, dt = \sqrt{R}\int_0^1 Q(t < \tau, \text{active at } t)\, dt \ge (1/2 - 1/8)\sqrt{R}.$$

For the terminal cost, let j be the last visited stage. The stopped variables satisfy $|M| \le b_j$, $Y \le H_j$, and $b_j \le \theta^{-1}(H_j)/H_j$. Convexity gives, pathwise,

$$\theta(|X|) \le \theta\big((Y/H_j)\theta^{-1}(H_j)\big) \le Y.$$

The stopped Y is a bounded positive martingale of mean one, so $\mathbb{E}\theta(|X|) \le 1$ and $\mathbb{E}|Y-1| \le 2$. Its integrand and entire state path are bounded, since only finitely many stages occur. Taking $L = \mathcal{L}\mathbf{1}_{t<\tau}$ and then A sufficiently large proves (L3). □

6.2 A concrete affine-orthogonal bubble

Lemma 6.3. *On $T = \{(x,y) : x, y \ge 0,\ x + y \le 1\}$ there is a nonzero continuous piecewise affine function b, compactly supported in its interior, such that $\int_T b\ell = 0$ for every affine function ℓ. For a uniform point U of T, the law of $c_b^{-1}\nabla b(U)$ is bounded, centered, has covariance I, and is invariant under all signed coordinate permutations. Moreover b is affine on a fixed subdivision into congruent right-isosceles triangles of a dyadic leg 2^{-k_0}.*

Proof. Take four centres $c_{ij} = (1/8 + i/8, 1/8 + j/8)$, $i, j \in \{0,1\}$, and radius $r = 1/64$. The four squares of radius r about these centres are disjoint and lie strictly inside T. Put

$$b(z) = \sum_{i,j=0}^{1} (-1)^{i+j}(r - \|z - c_{ij}\|_\infty)_+.$$

All four tents have equal mass. Their first moments are that mass times their centres. The checkerboard signs cancel the total mass and both first moments. Each tent's gradient distribution is supported on $\{0, \pm e_1, \pm e_2\}$ and is invariant under signed coordinate permutations; changing the sign of a tent preserves this law. Thus its covariance is $c_b^2 I$ for a fixed $c_b > 0$, and its mean is zero. All break lines have dyadic coordinates and directions horizontal, vertical, or slope ± 1. Choose a sufficiently fine dyadic square lattice through all centres and tent boundaries. In each small square choose the diagonal that contains its break line, if any; otherwise choose either diagonal. The resulting triangles, including the halves along the diagonal boundary of T, are congruent right-isosceles triangles, and b is affine on each. This proves every assertion. □

6.3 Bounded-state spatial realization

Here $\Psi : \mathbb{R}^2 \to [0,\infty)$ is any finite convex function with $\Psi(0) = 0$. We record carefully the bounded-state extension of the Gaussian bubble construction described below. It prevents any assumption about Gaussian integrability of θ from entering the argument. Let $D = (0,1)^2$, let W be white noise on D, and let P_m be averaging on its dyadic squares of side 2^{-m}.

Lemma 6.4 (Bounded-state bubble realization)**.** *Let L be a bounded predictable upper-triangular process with nonnegative diagonal entries, and suppose $Z_t = \int_0^t L_s\,dB_s$ stays in a fixed compact set. For every $\delta > 0$ there are random mean-zero Lipschitz potentials u_m such that, for every realization of their Gaussian signs,*

$$\int_D \Psi(\nabla u_m) \le \mathbb{E}\Psi(Z_1) + \delta + o(1), \tag{6.1}$$

$$\liminf_{m\to\infty} \frac{\mathbb{E}W(P_m u_m)}{\sqrt{m}} \ge c_* \left(\mathbb{E}\int_0^1 \sqrt{\det L_t}\,dt - \delta \right). \tag{6.2}$$

Here $c_* > 0$ is universal. The error and the uniform bound on ∇u_m may depend on the chosen control and on δ.

Proof. Compact-state approximation. Suppose $|Z_t| \le H$. Approximate L in $L^2(\mathbb{P}\otimes dt)$ by uniformly bounded upper-triangular simple processes V, constant on a finite deterministic partition and with block matrices bounded continuous functions of finitely many earlier Brownian increments. Density of predictable simple processes, followed by finite-dimensional continuous approximation, gives this assertion. Clip their diagonal entries to a fixed nonnegative interval, clip the off-diagonal entry, and add εI. The approximation still converges, and each fixed V has compact, uniformly invertible matrix range.

Write $Z^V = \int V\,dB$, and stop it at its first exit τ_V from the ball of radius $H+1$. Doob's inequality gives

$$\mathbb{P}(\tau_V \le 1) \le 4\mathbb{E}\int_0^1 |V_t - L_t|^2 dt.$$

The stopped states stay in this fixed ball. Uniform continuity of Ψ on the ball and convergence in probability give convergence of their terminal expected costs to $\mathbb{E}\Psi(Z_1)$. The rewards also converge: $\sqrt{\det V}$ converges in mean by uniform continuity on the bounded matrix range, and stopping loses at most a constant times $\mathbb{P}(\tau_V \le 1)$. Fix a sufficiently accurate V so both errors are less than δ. No cost outside a compact state set has been used.

Bounded innovations and stopping. Use the reference piecewise affine bubble b on a right-isosceles triangle T from Lemma 6.3. It is compactly supported in T, satisfies $\int_T b\ell = 0$ for every affine ℓ, and, for a uniform point $U \in T$, $\xi = c_b^{-1}\nabla b(U)$ is bounded, centered, has covariance I, and is invariant under signs and signed coordinate permutations. It is affine on a fixed congruent triangulation of depth k_0. Take $n = \lfloor m/(4k_0)\rfloor$ microsteps, evaluate the block matrices V_j from completed driver sums, and consider the ideal innovation martingale with increments $V_j\xi_k/\sqrt{n}$. Stop it after the first microstep whose state has norm at least $H+1$. This is a predictable decision for subsequent increments; the overshoot is $O_V(n^{-1/2})$.

The invariance principle and continuity of the finitely many coefficient maps give joint uniform-path convergence of the driver and the unstopped martingale, block by block. Their sphere-exit stopping times and stopped paths converge too. Indeed, within each fixed block the limiting diffusion matrix is constant and uniformly invertible, so its normal Brownian component crosses the sphere immediately upon first hitting; hitting exactly at a deterministic block endpoint has probability zero. These facts give almost sure continuity of the stopping map. All stopped states lie in a fixed compact set, hence their expected Ψ costs converge without a growth assumption. The bounded reward sums converge to $\mathbb{E}\int_0^{1\wedge\tau_V} \sqrt{\det V_t}\,dt$.

The spatial branch law and its pathwise cost. At block j, put $a_j = \sqrt{\det V_j}$ and choose $A_j^{-T} = V_j/a_j$, so $\det A_j = 1$. On an active triangle $Q = x_Q + d_Q A_j O_Q T$, add the signed bubble

$$b_Q(x) = \frac{d_Q a_j}{c_b\sqrt{n}} b\big(d_Q^{-1} O_Q^T A_j^{-1}(x - x_Q)\big), \qquad s_Q = \operatorname{sgn} W(b_Q). \tag{L8}$$

Its gradient increment is $V_j\xi_Q/\sqrt{n}$, with $\xi_Q = s_Q c_b^{-1} O_Q \nabla b$. Before the first microstep, tile the square in the initial chart A_1 at mesh $2^{-\lfloor\sqrt{m}\rfloor}$, retaining complete triangles and freezing the boundary remainder at gradient zero. Mark a parent inactive after its state reaches the barrier; this requires no retiling. At each later fixed control-block boundary retile the surviving triangles in the new charts, at relative mesh $2^{-\lfloor\sqrt{m}\rfloor}$, retaining complete triangles and discarding the boundary remainder. Compact chart bounds give total discarded area $o_V(1)$, and minimum intrinsic leg $d_Q \geq 2^{-m/3}$, by the boundary-strip estimate for a triangle under two uniformly bounded charts.

Freeze *any* assignment of all bubble signs and follow a uniform spatial point. Conditional on its parent and all previous child choices, the point is uniform in that parent. Congruent subdivision therefore gives the same law of ξ_Q at every step: the fixed sign and signed permutation do not change that law. At a retiling, conditional on the retained new triangle, uniformity holds again. After geometric loss append independent artificial innovations. After the state barrier do the same for the driver, leaving the gradient fixed. Induction proves that the ideal innovations are independent with law ξ, for every frozen sign assignment.

The actual spatial gradient is precisely this ideal state-stopped martingale further stopped at geometric loss. The filtration records only parent and child identities and the predictable retiling decisions, not the location inside a current triangle. Thus conditional Jensen at that stopping time gives

$$\int_D \Psi(\nabla u_m) \leq \mathbb{E}_{\text{ideal}} \Psi(\text{terminal stopped chain}). \tag{L9}$$

The right side is independent of all frozen Gaussian signs. This is a pathwise bound, not an estimate averaged over white noise. It proves (6.1). Discarded pieces freeze their old gradient, so the asserted compact gradient bound holds everywhere.

Fresh Gaussian rewards. Reveal bubbles breadth first. Every earlier bubble is disjoint from Q or affine on Q, so its pairing with b_Q vanishes by the affine moments of b. Conditional on earlier Gaussian observations, $W(b_Q)/\|b_Q\|_2$ is consequently a fresh standard normal. More explicitly, the directions take finitely many values at each fixed depth; induction on this finite decision tree leaves an isonormal Gaussian process on the orthogonal complement of the already revealed directions. For any past-measurable test v, decomposition into its b_Q component and the remaining Gaussian and past-measurable components yields

$$\mathbb{E}[s_Q W(v) \mid \text{past}] = \sqrt{2/\pi}\, \frac{\langle v, b_Q\rangle}{\|b_Q\|_2}.$$

In particular, with $v = P_m b_Q$, the numerator is $\|P_m b_Q\|_2^2$. Poincaré on dyadic squares gives $\|P_m b_Q - b_Q\|_2 \leq C_V 2^{-m} d_Q^{-1} \|b_Q\|_2 = o(1)\|b_Q\|_2$. Determinant-one scaling in (L8) gives $\|b_Q\|_2 = c_b' a_j |Q|/\sqrt{n}$, with universal $c_b' > 0$. Thus each active parent has conditional expected reward at least $c a_j |Q|/\sqrt{n}$. At each microstep their total coefficient is the spatial expectation of a_j before the state barrier, up to $o_V(1)$ geometric loss. Summing, using the stopped-chain convergence and $n \asymp m$, proves (6.2). □

6.4 Necessary conditions for two arbitrary Young functions

Let θ_1, θ_2 be finite convex Young functions with $\theta_i(t) \geq t$, and put

$$r_i(S) = \frac{e^S}{\theta_i^{-1}(e^S)}, \qquad S_j = 2^j \quad (j \geq 1). \tag{P1}$$

These functions are nondecreasing, since $\theta_i^{-1}(u)/u$ is nonincreasing. The inverse-product inequality proved above gives $r_i(S) \asymp (\theta_i^*)^{-1}(e^S)$, with absolute comparison constants.

Lemma 6.5 (Finite weighted entropy)**.** *Fix $0 < \gamma < e^{-1}$. For finitely many positive weights v_j, set*

$$F(v) = \sup_{\substack{q_j \geq 0 \\ \sum q_j \leq \gamma}} \sum_j q_j \sqrt{\log(1/q_j)}\, v_j.$$

Let $T(v) > 0$ be determined by $\sum_j \exp[-T(v)^2/v_j^2] = e^{-1}$. Then

$$\gamma T(v) \leq F(v) \leq C_\gamma T(v), \qquad T(v) \geq \max_j v_j. \tag{P2}$$

Consequently, with $v_j = \sqrt{S_j}/r_1(S_j)$, a uniform bound $F((v_j)_{j\leq J}) \leq C r_2(S_J)$ is equivalent to the existence of $c > 0$ such that, for every J,

$$\sum_{j \leq J} \exp\left[-c \frac{r_1(S_j)^2 r_2(S_J)^2}{S_j}\right] \leq e^{-1}. \tag{P3}$$

Proof. The lower bound follows by taking $q_j = \gamma e \exp[-T^2/v_j^2]$: these probabilities sum to γ and $\log(1/q_j) \geq T^2/v_j^2$. Also each term of the defining sum for T is at most e^{-1}, giving $T \geq \max v_j$. For the upper bound put $a_j = T/v_j \geq 1$. Split the normalized entropy sum according as $\sqrt{\log(1/q_j)} \leq 2a_j$ or not. The first part is at most 2γ, while the second is at most $C \sum_j e^{-2a_j^2} \leq Ce^{-1}$, exactly as in Lemma 6.1. The last assertion follows from the monotonicity in T of its defining exponential sum. We will also use $T(v) \asymp \mathbb{E} \max_j v_j |G_j|$ for independent standard normals. For the upper bound use the Gaussian union bound at levels sT, $s \geq 2$, and $T^2/v_j^2 \geq 1$, then integrate the tail. For the lower bound, the elementary bound $\mathbb{P}(|G| \geq x) \geq ce^{-x^2}$, together with independence, gives a fixed positive probability that the maximum exceeds $T/2$. □

Proposition 6.6 (Two necessary prefix conditions)**.** *If the estimate in the question holds uniformly in the grid size and sampling parameter, then* (P3) *holds for some $c > 0$, and so does its axis-interchanged version*

$$\sum_{j \leq J} \exp\left[-c \frac{r_2(S_j)^2 r_1(S_J)^2}{S_j}\right] \leq e^{-1} \qquad (J \geq 1). \tag{P4}$$

The constant c may be enlarged to be the same in both conditions.

Proof. We prove the assertion by a bounded-state martingale construction. The cost is now $\Psi(x, y) = \theta_1(|x|) + \theta_2(|y|)$. The proof of Lemma 6.4 applies to this cost without change: its cost argument uses only convexity and continuity on a fixed compact set. Thus arbitrarily large determinant reward at a uniformly bounded expected terminal Ψ cost contradicts the desired Gaussian, and hence empirical, estimate.

For clarity write

$$v_j = \frac{\sqrt{S_j}}{r_1(8S_j)}, \qquad F_J = \sup_{\substack{q_j \geq 0 \\ \sum_{j_0 \leq j \leq J} q_j \leq \gamma}} \sum_{j=j_0}^{J} q_j \sqrt{\log(1/q_j)}\, v_j.$$

We show that $F_J/r_2(8S_J)$ must stay bounded. Replacing $8S_j$ by S_j merely shifts indices by three and multiplies the weights by $\sqrt{8}$. Omitting the fixed initial segment is harmless: its entropy functional is finite, the functional is subadditive over index sets, and $r_2 \geq 1$. Lemma 6.5 then gives precisely (P3).

Choose probabilities within a factor two of the supremum defining F_J. Their total mass can be taken to equal γ, since $q\sqrt{\log(1/q)}$ is increasing for $q \le \gamma$, with γ chosen sufficiently small. Put

$$y_0 = \frac{1}{r_2(8S_J)}, \quad H_j = e^{8S_j}, \quad b_j = \frac{\theta_1^{-1}(H_j)}{y_0 H_j} = \frac{v_j}{y_0\sqrt{S_j}}, \quad \ell_j = q_j b_j, \quad K_j = \log(1+q_j^{-2}). \tag{P5}$$

Use the operational local-time stages of Lemma 6.2, now with $w_j(v) = S_j/[K_j(v+\ell_j^2)]$. Their proofs were scale invariant in b_j, so they give

$$\mathbb{P}(\text{failure in stage } j) \le Cq_j, \quad \mathbb{E}[R_j; \text{success}] \ge \frac{cq_j\sqrt{K_j}v_j}{y_0}, \quad R_j \le \frac{2v_j}{y_0\sqrt{K_j}}. \tag{P6}$$

Condition on all the finitely many stages succeeding. The resources are independent and their sum has mean $\mu \ge cF_J/y_0$. On the other hand

$$F_J \ge \gamma\sqrt{\log(1/\gamma)} \max_{j_0 \le j \le J} v_j,$$

by placing all the mass on a largest weight. Hence the deterministic maximum of the individual R_j's is at most $C_\gamma F_J/y_0 \le C'_\gamma \mu$. Their conditional variance is at most this maximum times μ. Paley–Zygmund therefore gives a constant $p_\gamma > 0$, independent of J and the two profiles, such that with probability at least p_γ all stages succeed and their total resource exceeds $\mu/4$. Here we used $\mathbb{P}(\text{all successes}) \ge 1 - C\gamma \ge 1/2$.

Set $R = \mu/4$ and embed the stages in ordinary time by $dt = \sqrt{w_j(v)}\,dv/R$. As before, $\sigma\beta = R$, $dM = \sigma\, dB^1$, and the multiplicative variance clock up to stage j is at most $2S_j$. This time start the multiplicative martingale at y_0:

$$Y_t = y_0 \exp\left(\int_0^t \beta_r\, dB_r^2 - s_t/2\right), \qquad X_t = Y_t M_t.$$

Before the additional cap, β is still a function of B^1 alone. Consequently the uncapped measure $dQ = (Y_1/y_0)d\mathbb{P}$ preserves its entire B^1 marginal. Stop at the first crossing $Y = y_0 H_j$ during stage j. Under Q, $\log(Y/y_0)$ is Brownian motion plus half its variance clock, conditionally on B^1. The cap probability is therefore at most $\sum_{j \ge j_0} 2e^{-49S_j/4}$. Choose j_0 once so that this is less than $p_\gamma/2$. The stopped martingale $(X, Y - y_0)$ has triangular integrand $\left(\begin{smallmatrix} Y\sigma & M Y\beta \\ 0 & Y\beta \end{smallmatrix}\right)$ before stopping. Its expected determinant reward is at least

$$\frac{p_\gamma}{2} y_0\sqrt{R} \ge c_\gamma\sqrt{y_0 F_J}. \tag{P7}$$

The expected terminal cost stays bounded, independently of J. Indeed, at a terminal state in stage j, convexity and (P5) give

$$\theta_1(|X|) \le \theta_1\left(\frac{Y}{y_0 H_j}\theta_1^{-1}(H_j)\right) \le Y/y_0.$$

Also $Y \le y_0 H_J = \theta_2^{-1}(H_J)$, whence $\theta_2(Y) \le Y/y_0$. Since the stopped Y is a bounded martingale of mean $y_0 \le 1$,

$$\mathbb{E}\{\theta_1(|X|) + \theta_2(|Y - y_0|)\} \le \mathbb{E}\theta_1(|X|) + \mathbb{E}\theta_2(Y) + \theta_2(y_0) \le 2 + \theta_2(1). \tag{P8}$$

The first inequality uses $|Y - y_0| \le \max(Y, y_0)$. Put $C_0 = 3 + \theta_2(1)$. Apply the spatial lemma with a fixed error less than $1/2$. For all sufficiently fine meshes its pathwise energy is at most C_0,

independently of J. Dividing the potential by C_0 gives energy at most one by convexity, and divides its reward by only this fixed constant. For every fixed finite list the state and the integrand are bounded; in particular all compact-state approximation and pathwise spatial cost arguments apply. If $F_J/r_2(8S_J) = y_0F_J$ were unbounded, (P7)–(P8) would contradict the desired estimate. This proves (P3). Interchanging the axes proves (P4). □

Remark 6.7 (A simpler pointwise necessary condition)**.** *The full uniform family of prefix sums implies*

$$r_1(S)r_2(S) \geq c\sqrt{S} \qquad (S \text{ sufficiently large}). \tag{P9}$$

To see the needed uniform lower bound, put $p_J = r_1(S_J)^2 r_2(S_J)^2/S_J$. If p_J is small, then for $k \leq J$ monotonicity gives $r_1(S_k)^2 r_2(S_J)^2/S_k \leq 2^{J-k}p_J$. Thus a block of order $\log(1/p_J)$ preceding terms in the first prefix sum are bounded below by e^{-c}, contradicting its uniform boundedness. There are enough preceding terms because both inverse profiles have fixed positive lower bounds. Dyadic values imply the full statement by monotonicity. Equivalently, $\theta_1^{-1}(t)\theta_2^{-1}(t) \leq Ct^2/\sqrt{\log t}$. The prefix conditions can be strictly stronger: the examples below have both costs superlinear and satisfy this pointwise condition while the desired estimate fails.

6.5 An exact subclass with comparable power profiles

Proposition 6.8 (Logarithmic power costs)**.** *If $\theta_i(t) \asymp t(\log t)^{\beta_i}$ at infinity, with $\beta_i \geq 0$ and Young normalization near zero, the bound holds exactly when*

$$\beta_1 + \beta_2 > \tfrac{1}{2}, \quad \text{or} \quad \beta_1 + \beta_2 = \tfrac{1}{2} \text{ and } \beta_1\beta_2 > 0.$$

Proof. Here $r_i(S) \asymp S^{\beta_i}$. In the first orientation the prefix exponents are comparable to $2^{(2\beta_1-1)j+2\beta_2 J}$. If $\beta_1 < 1/2$ and the sum is at least $1/2$, they grow geometrically as $J - j$ increases; at equality the rate is positive exactly when $\beta_2 > 0$. If $\beta_1 = 1/2$ and $\beta_2 > 0$, the sum is at most $J\exp(-c2^{2\beta_2 J})$. If $\beta_1 > 1/2$, the sum is bounded by the convergent infinite series $\sum_j \exp(-c2^{(2\beta_1-1)j})$. Apply the same argument after interchanging coordinates. If the sum of exponents is smaller than $1/2$, the pointwise product condition fails. At $(\beta_1, \beta_2) = (1/2, 0)$ the first entropy weights are constant, and their maximum grows as $\sqrt{\log J}$, so the corresponding prefix condition fails. The other endpoint is identical with the coordinates interchanged. □

Proposition 6.9. *Suppose that for some fixed $\kappa > 0$, $r_1(S) \asymp r_2(S)^\kappa$ for all sufficiently large S. Then the desired estimate holds if and only if* (P9) *holds. In particular, if $\theta_1 = \theta_2 = \theta$, its exact condition is*

$$\theta(t) \geq c\,t(\log t)^{1/4} \qquad (t \text{ sufficiently large}). \tag{P10}$$

Proof. Necessity follows from Proposition 6.6. Conversely, power comparability and (P9) imply

$$r_1(S) \geq cS^{\beta_1}, \quad r_2(S) \geq cS^{\beta_2}, \qquad \beta_1 = \frac{\kappa}{2(\kappa+1)}, \quad \beta_2 = \frac{1}{2(\kappa+1)}.$$

Both exponents are positive and sum to $1/2$. These lower bounds imply the two prefix conditions directly: the first normalized inverse product is at least $c2^{\beta_2(J-j)}$ and its axis-swapped counterpart is at least $c2^{\beta_1(J-j)}$. Both exponential sums are bounded geometric-tail sums. The general prefix criterion therefore proves sufficiency. Equivalently the gradient costs dominate $t(\log t)^{\beta_i}$.

When the costs are equal, $\kappa = 1$ and the inverse condition is $r(S) \geq cS^{1/4}$. For any fixed $\beta > 0$, $r(S) \geq cS^\beta$ is equivalent to $\theta(t) \geq c't(\log t)^\beta$ at infinity. For one direction put $S = \log\theta(t) \geq \log t$. For the other put $t = \theta^{-1}(e^S)$: if $t \leq e^{S/2}$, then $e^S/t \geq e^{S/2}$; otherwise $\log t \geq S/2$, and the stated cost lower bound gives $e^S/t \geq c'(S/2)^\beta$. This proves (P10). □

6.6 An iterated-logarithm family near the hard endpoint

The following calculation separates the pointwise product obstruction from the extra entropy obstruction. All tail comparisons are for sufficiently large arguments; functions may be patched near zero to be Young functions without changing the conclusions.

Proposition 6.10. *Suppose*

$$\theta_1(t) \asymp t(\log t)^{1/2}(\log\log t)^{\gamma_1}(\log\log\log t)^{\delta_1},$$
$$\theta_2(t) \asymp t(\log\log t)^{\gamma_2}(\log\log\log t)^{\delta_2},$$

where either $\gamma_2 > 0$, or $\gamma_2 = 0$ and $\delta_2 \geq 0$, as required for the second cost to dominate t. Then the desired estimate holds exactly when

$$\gamma_1 + \gamma_2 > 0, \quad \text{or} \quad \gamma_1 + \gamma_2 = 0 \ \text{ and } \ \delta_1 + \delta_2 \geq \tfrac{1}{2}. \tag{P11}$$

The two prefix conditions, and hence the general theorem, give this necessary and sufficient condition. In particular, for every $a > 0$, the two superlinear costs

$$\theta_1(t) \asymp \frac{t\sqrt{\log t}}{(\log\log t)^a}, \qquad \theta_2(t) \asymp t(\log\log t)^a$$

fail the desired estimate, although their product has the pointwise critical order $t^2\sqrt{\log t}$.

Proof. Inversion of these iterated-logarithm functions gives

$$r_1(S_j) \asymp 2^{j/2} j^{\gamma_1}(\log j)^{\delta_1}, \qquad r_2(S_j) \asymp j^{\gamma_2}(\log j)^{\delta_2}.$$

Thus the first entropy weights are $v_j \asymp j^{-\gamma_1}(\log j)^{-\delta_1}$. For independent standard normals G_j, their entropy functional is comparable to $\mathbb{E}\max_{j\leq J} v_j|G_j|$, or directly to the exponential-sum threshold in Lemma 6.5. The following elementary estimates therefore suffice:

$$F_J \asymp \begin{cases} J^{-\gamma_1}(\log J)^{1/2-\delta_1}, & \gamma_1 < 0, \\ 1 + (\log J)^{\max(1/2-\delta_1,0)}, & \gamma_1 = 0, \\ 1, & \gamma_1 > 0. \end{cases} \tag{P12}$$

For increasing polynomial weights the last $J/2$ indices give the lower bound, and the usual Gaussian maximum bound gives the upper bound. Polynomially decaying weights have a summable inverse-square exponential series. For pure logarithmic weights with $\delta_1 < 1/2$, put $T = C(\log J)^{1/2-\delta_1}$; for $j \leq J$, $T^2(\log j)^{2\delta_1} \geq C^2 \log j$ when $0 \leq \delta_1 < 1/2$, so the exponential-sum upper bound follows. Negative δ_1 is covered by the increasing-weight argument. If $\delta_1 \geq 1/2$, the infinite exponential series converges for a sufficiently large constant. These observations prove (P12).

The swapped weights are $\sqrt{S_j}/r_2(S_j) \asymp 2^{j/2} j^{-\gamma_2}(\log j)^{-\delta_2}$. Since they grow geometrically, their entropy functional is comparable to the final weight. Consequently the two necessary prefix bounds are equivalent, in this family, to (P11). Proposition 6.6 proves failure in every excluded case.

Conversely the same entropy calculation verifies both prefix conditions in every included case. The general two-function criterion proves the matching upper bound, including equality in (P11). □

Remark 6.11. *The examples above are consequences of the general two-function criterion; they impose no additional regularity beyond the finite Young function hypotheses. The explicit iterated-logarithm comparisons only make that criterion easier to read in this subclass.*

7 Completion of the Young-function theorem

The prefix conditions give the allocated phase and the exact Gaussian current of Proposition 5.1. For every potential f of continuum modular energy at most one, weak integration by parts and the complementary Young inequality yield

$$|W_h^0(f)| \le 2\sum_{i=1}^{2} \|J_i\|_{L^{\psi_i}} \|\partial_i f\|_{L^{\theta_i}} \le 4 \max_i \|J_i\|_{L^{\psi_i}}.$$

The grid–continuum comparison therefore proves the Gaussian upper bound. Conversely, if either prefix condition fails, the bounded-cost local-time controls and bounded-state bubble realization give deterministic-energy potentials with arbitrarily large multiples of $\sqrt{p}$ as expected pairing. Cell averaging transfers these to the lattice without additional energy. This proves the converse Gaussian implication. The upper empirical comparison and the uniform-law Skellam converse prove the equivalence with (17.18), including $m_0 \asymp p$. The finitely many coarse meshes omitted during the constructions are absorbed by enlarging the fixed constant. This proves Theorem 1.1. □

8 Finite Borel costs and quantitative convexification

For $j = 1, 2$, let $\theta_j : [0, \infty) \to [0, \infty)$ be finite Borel functions satisfying $\theta_j(t) \ge t$. Let ϕ_j be the even convex envelope of $s \mapsto \theta_j(|s|)$, and put

$$a_j = \inf \theta_j = \phi_j(0), \quad a = a_1 + a_2, \qquad \vartheta_j(t) = \phi_j(t) - a_j + t \quad (t \ge 0).$$

The equality at zero follows by mixing s and $-s$. Since $\phi_j(s) \ge |s|$, the functions ϑ_j are finite convex Young functions, vanish at zero, and dominate t.

Theorem 8.1. *If $a < 1$, the original pair satisfies the discrepancy bound if and only if the normalized Young pair $(\vartheta_1, \vartheta_2)$ satisfies the criterion proved above. If $a \ge 1$, the original pair always satisfies the discrepancy bound.*

We first give the elementary finite local lemma needed to handle costs that are unbounded on every neighborhood.

Lemma 8.2 (Finite local lemma)**.** *Suppose $E_1, \dots, E_N$ are events determined by independent variables, and each event shares variables with at most d others. If, for some $0 < x < 1$, $\mathbb{P}(E_i) \le x(1-x)^d$ for every i, then*

$$\mathbb{P}\Big(\bigcap_i E_i^c\Big) \ge (1-x)^N > 0.$$

Proof. Inductively prove that, for every S not containing i, $\mathbb{P}(E_i \mid \bigcap_{j\in S} E_j^c) \le x$. Partition S into the neighbors A of i and the remaining indices B. The event E_i is independent of the entire family indexed by B. Its conditional probability after also excluding the events in A is therefore at most

$$\frac{\mathbb{P}(E_i)}{\mathbb{P}(\bigcap_{j\in A} E_j^c \mid \bigcap_{j\in B} E_j^c)} \le \frac{x(1-x)^d}{(1-x)^{|A|}} \le x.$$

The middle bound follows by the induction hypothesis and the chain rule; its conditioning sets have smaller cardinality than S. The same chain rule now gives the asserted lower bound. □

Proof of Theorem 8.1. For sufficiency when $a < 1$, note that $\vartheta_j(t) \le \phi_j(t) + t \le 2\theta_j(t)$. Dividing an original admissible function by two puts it in the normalized Young ball. We prove the converse by transferring the bounded-gradient Gaussian lower tests from the Young-pair theorem.

Fix arbitrarily large D. That lower construction, followed by a fixed rescaling, provides $H < \infty$ and, for arbitrarily large m, random functions u_m on the unit square with

$$\int u_m = 0, \quad |\partial_j u_m| \le H, \quad \int \sum_j \vartheta_j(|\partial_j u_m|) \le \eta, \qquad \mathbb{E}\langle P_{2^{-m}} W^0, u_m \rangle \ge D\sqrt{m}.$$

Here $\eta > 0$ is fixed and sufficiently small relative to $1 - a$. Their convex-envelope energy is at most $a + \eta$. Conditional expectation given the fixed coarse histogram preserves these pathwise convex energy and gradient bounds and preserves the coarse reward. Indeed, take conditional expectations on a countable dense set of spatial points; the uniform Lipschitz bound gives a continuous version. Fubini in the weak derivative identity identifies its derivatives with the conditional expectations of the original derivatives, and Jensen gives the pathwise energy bound. Hence assume u_m is measurable with respect to $P_{2^{-m}} W^0$.

Finite base gradients and finite well slopes. Extend u_m evenly by reflection to the period-two torus and convolve with a coordinatewise-even nonnegative probability kernel at scale $\rho = 2^{-2m}$. The result u_m^ρ has coordinate gradients bounded by H, Hessian bounded by CH/ρ, and uniform error at most $CH\rho$. Jensen and reflection symmetry show that its convex-envelope energy on the square has not increased.

Triangulate the square by a grid of side $\delta = 2^{-3m}$ and a fixed right-triangle diagonal. Round the values of u_m^ρ at vertices to the nearest multiple of $q\delta$, and interpolate affinely to obtain $\widehat{u}_m$. Here $q > 0$ is fixed and small. The interpolation estimates are

$$\|\widehat{u}_m - u_m^\rho\|_\infty \le C(H + q)\delta, \qquad |\nabla \widehat{u}_m - \nabla u_m^\rho| \le CH\delta/\rho + Cq.$$

Its gradients therefore lie in a fixed finite subset of $q\mathbb{Z}^2$, independent of m. Continuity of the convex ϕ_j on compact intervals permits choosing q, and then m, so that its convex-envelope energy is at most $a + 2\eta$. To justify the finite-well assertion, the infimum of the cost of all finite convex combinations with prescribed mean is a finite convex minorant, hence equals ϕ_j. In any one approximating finite combination, minimizing the linear cost under its mass and mean constraints has an extreme solution supported at at most two points. For each coordinate of each base gradient, choose such a two-point original-well mixture with that mean and cost at most the corresponding ϕ_j value plus η. Only finitely many well slopes are needed, and their original costs are finite.

Two clipped waves. On a base triangle with gradient (b_1, b_2), add a nonnegative periodic triangular wave in x, of amplitude at most $\epsilon_1 = 2^{-4m}$, whose derivative changes b_1 to its two chosen first-coordinate wells in the prescribed proportions. Clip this wave by the distance to the triangle boundary. The correction vanishes at the boundary, is continuous and piecewise affine, and has bounded gradients. Outside a boundary strip of width $C\epsilon_1$, its first derivative is an original well and its second derivative remains b_2.

On every retained affine polygon, add a corresponding triangular wave in y, of amplitude at most $\epsilon_2 = 2^{-5m}$, clipped by distance to that polygon's boundary. Keep the first boundary strips unchanged. Denote the resulting continuous piecewise affine function by v_m. It agrees with $\widehat{u}_m$ on all base triangle boundaries, has gradient bounded by a fixed K, and

$$\|v_m - \widehat{u}_m\|_\infty \le \epsilon_1 + \epsilon_2.$$

Call the interiors where both derivatives are original well slopes the good regions. The remaining region B_m satisfies

$$|B_m| \le C(\epsilon_1/\delta + \epsilon_2/\epsilon_1).$$

Indeed the total perimeter of the original triangles is $O(\delta^{-1})$ and that of the first retained polygons is $O(\epsilon_1^{-1})$. A polygon's inner boundary strip has area at most its perimeter times its width. The total interface length of the final affine partition is at most C/ϵ_2, and its number of vertices is at most $C/(\epsilon_1\epsilon_2)$. These bounds follow by slicing the original triangles into their first-wave stripes and each resulting polygon into second-wave stripes; clipping adds a bounded number of segments per stripe. More explicitly, there are $O((\delta\epsilon_1)^{-1})$ first polygons. Each has height at most $C\delta$; the second slicing therefore gives $O((\epsilon_1\epsilon_2)^{-1})$ pieces in total. All constants are uniform because the base gradients and well slopes come from fixed finite libraries.

Only well costs are needed on the good regions. Their summed integral is at most

$$a + 4\eta + C_{\text{well}}(\epsilon_1/\delta + \epsilon_2/\epsilon_1).$$

For clarity, the well proportions on a polygon differ from their prescribed means by at most the wave period times its perimeter: integrate by parts a bounded periodic primitive of the zero-mean stripe indicator. Summing the first and second wave errors gives the displayed bound. Removing the boundary strips cannot increase these nonnegative well costs. We make no assertion about the original cost of the continuum gradients in B_m.

A bounded-cost assignment on affected lattice edges. Take $h = 2^{-8m}$ and initially sample v_m at the centers of the h-squares. Declare a grid edge defective if its joining segment is not contained in a single good affine region. Mark both endpoints of every defective edge, and call all edges incident to a marked vertex affected. The normalized number of affected edges is at most

$$C\big(|B_m| + h/\epsilon_2 + h^2/(\epsilon_1\epsilon_2)\big) = o(1).$$

This is the area count for B_m, plus the standard grid-edge count along a polygonal interface; marking endpoints multiplies the count by only a numerical constant. To verify that count directly, cover each interface segment by a rectangle of width Ch and cover each endpoint by a disk of radius Ch. Their total area is at most Ch times the interface length plus Ch^2 times its number of vertices. Every defective grid edge lies in B_m or in this cover, and disjoint grid squares convert this area bound to the displayed normalized count.

Fix $\tau = 1/10$ and perturb each marked vertex value independently by $h\xi_v$, with ξ_v uniform on $[-\tau, \tau]$; keep other vertices fixed. Every affected edge has at least one random endpoint. Its derivative equals a fixed number in $[-K, K]$ plus the difference of at most two such uniforms, so it has density at most $1/(2\tau)$ on $[-K - 2\tau, K + 2\tau]$. Since the original costs are finite Borel functions, their superlevel sets on this compact interval have Lebesgue measures tending to zero. Choose one fixed M large enough that, for each direction, the probability its affected-edge cost exceeds M is at most $1/100$.

Each bad-cost event depends on at most the two endpoint jitters and shares variables with at most six other edge events. Since $1/100 < (1/8)(7/8)^6$, Lemma 8.2 supplies an assignment for which every affected-edge cost is at most M. No assignment can change any value by more than τh. For fixed m there are only finitely many rounded base vertex arrays, because $|u_m| \le 2H$. Choose a successful assignment once for each such array. The resulting assignment rule is measurable and depends only on the coarse histogram. This avoids any measurable-selection or algorithmic assumption.

Let z_m be the cellwise constant function given by these perturbed samples. Unaffected edges still see exact original-well derivatives. Their area-counting discrepancy is controlled by the same interface count above, and affected edges cost at most M. Consequently its normalized original lattice energy is at most

$$a + 4\eta + o(1).$$

Choose η with $a + 5\eta < 1$. For all sufficiently large m, z_m is original-cost admissible. Neither integrability nor local boundedness of the original costs was used.

The same proof also permits either original cost to be locally bounded without being measurable. On the fixed compact interval of affected slopes choose M above that cost's supremum. Its bad-edge events are then empty, so no measurability is needed for that coordinate. The other coordinate may independently be measurable or locally bounded. In particular this proves the stated classification for every locally bounded pair, with no regularity assumption beyond local boundedness.

The Gaussian reward and depth. The function z_m is coarse-histogram measurable. Added fine Gaussian bridges therefore have conditional mean zero, giving

$$\mathbb{E}\langle P_h W^0, z_m\rangle = \mathbb{E}\langle P_{2^{-m}} W^0, z_m\rangle.$$

Furthermore

$$\|z_m - u_m\|_\infty \le C(H+q)\rho + \epsilon_1 + \epsilon_2 + (K+\tau)h, \qquad \mathbb{E}\|P_{2^{-m}} W^0\|_1 \le C 2^m.$$

The expected reward is thus $D\sqrt{m} - o(1)$. Dividing the fine cell values by h gives a lattice function with budget h^{-2} and standard lattice Gaussian reward $h^{-2}(D\sqrt{m} - o(1))$. The new depth is $p = 8m$, a fixed factor. Since D is arbitrary, the Gaussian estimate fails for the original pair, and the Gaussian–empirical lower comparison transfers this failure to the sampling regime of the book.

Baseline degeneracies. If $a > 1$, the baseline $an(n-1)$ exceeds n^2 for all sufficiently large n; only finitely many sizes remain. Their bounds follow by finite dimensionality and the empirical–Gaussian upper comparison. If $a = 1$, admissibility and $\theta_j(t) \ge \max(a_j, t)$ imply

$$\sum_{j,e} (|D_j h| - a_j)_+ \le n.$$

The linear dyadic Gaussian divergence inverse $J = TG^0$ satisfies $\mathbb{E}\sum_e |J_e| \le Cn^2\sqrt{p}$ and $\mathbb{E}\|J\|_\infty \le Cp$. Summation by parts gives

$$|\langle G^0, h\rangle| \le \sum_j a_j \sum_{e\|j} |J_e| + n\|J\|_\infty.$$

Its expectation is at most $Cn^2\sqrt{p} + Cnp \le Cn^2\sqrt{p}$. The empirical upper comparison proves the asserted estimate. □

9 Why completely nonmeasurable costs are different

9.1 A genuine obstruction to unrestricted two-cost convexification

For two completely arbitrary finite functions, replacing the costs by their even convex envelopes is not a valid general principle. The following example uses functions that are not measurable; the literal formulation $\theta_j(t) \ge t$ does not exclude them.

Proposition 9.1. *There are finite functions $\theta_1, \theta_2 : [0,\infty) \to [0,\infty)$ with $\theta_j(0) = 0$ and $\theta_j(t) \geq t$ such that:*

1. *both even convex envelopes are exactly $t \mapsto |t|$;*
2. *the desired $O(n^2\sqrt{pm_0})$ empirical discrepancy estimate holds for the pair (θ_1, θ_2).*

Thus the unrestricted pair problem cannot be classified solely by these pointwise convex envelopes, since the linear–linear pair has already been shown to fail the estimate.

Proof. Choose a Hamel basis of $\mathbb{R}$ over $\mathbb{Q}$ containing 1 and $\sqrt{2}$. Let $P : \mathbb{R} \to \mathbb{Q}$ extract the coefficient of 1. It is additive, $P^2 = P$, and both its image and kernel are dense: they contain $\mathbb{Q}$ and $\sqrt{2}\mathbb{Q}$, respectively. Set, for $t \geq 0$,

$$\theta_1(t) = t + |P(t)|^2, \qquad \theta_2(t) = t + |t - P(t)|^2.$$

These functions are finite and dominate t. The first equals t on a dense subset, as does the second. Every finite convex minorant of either even cost is continuous, hence is at most $|t|$ by density; $|t|$ itself is a minorant. This proves the first assertion.

Let h satisfy the original budget n^2, and write $a(v) = P(h(v))$, $b(v) = h(v) - a(v)$. Additivity of P, and its oddness, give the exact identity

$$\mathrm{TV}(h) + \sum |D_x a|^2 + \sum |D_y b|^2 \leq n^2.$$

There are fewer than n^2 edges in each direction. Cauchy–Schwarz therefore gives

$$\sum |D_x a| \leq n^2, \qquad \sum |D_y b| \leq n^2,$$

and the identity $h = a + b$ implies

$$\sum |D_y a| \leq \sum |D_y h| + \sum |D_y b| \leq 2n^2, \qquad \sum |D_x b| \leq 2n^2.$$

Consequently a lies in four times the anisotropic Young ball with costs $(t + t^2, t)$, while b lies in four times the ball with costs $(t, t + t^2)$. Both balls satisfy the desired estimate by Theorem 1.1: the linear complementary width is constant, whereas the quadratic-cost complementary width grows exponentially in the log-level. The resulting prefix sums are bounded.

The centered empirical functional is linear in the test function, so its value at $h = a + b$ is bounded by the sum of its absolute values at a and b. Taking the two suprema and expectations proves the second assertion, with a universal constant. The same argument works directly for Gaussian width. □

Finite Borel costs, including those unbounded on compact sets, are covered by the preceding quantitative relaxation theorem. This example lies outside that measurable setting and shows that the qualification cannot simply be deleted from a convex-envelope classification.

Provenance of the methods. The discrepancy problem and its normalization are from Talagrand's second edition, Chapter 17. The local Gaussian current and adaptive bubble constructions were developed in the Ultimatematching project, in particular the Goldman–Grotto–Trevisan source files `umt.tex` and `endpoints.tex`. All ingredients used here are proved in this note. The two prefix conditions, width allocation, rounded phase, two-cost repair estimate, and measurable-cost lattice relaxation are the extensions developed in this review.

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